



Research paper

A study on a physical based manoeuvring mathematical model for submarines

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ABSTRACT

The manoeuvrability of submarines is one of the important factors from mission and safety points of view. In particular, manoeuvring motions with a large drift angle and/or angle of attack in both the horizontal and the vertical planes have different motion parameters compared to the normal manoeuvring motions. To evaluate the manoeuvrability during these types of particular motions of submarines, especially at the initial design stage, the manoeuvring hydrodynamic derivatives that are related to various manoeuvring motions must be identified.

Karasuno model, which is termed a physical based model, has the advantages that it needs only model experiments for drift motion and it can provide greater accuracy under motion with large angles. Despite these advantages, this model has not been applied to other vessels excluding fishing vessels. The applicability of this method to submarines therefore should be examined. The applicability of Karasuno model for submarines is validated through a computational fluid dynamics (CFD) analysis and comparisons with experimental data in this work. The findings contribute to identifying the manoeuvring hydrodynamic forces for submarines and evaluating their manoeuvring performance. Through rigorous validation, the potential of the proposed model as a reliable tool for predicting and analyzing submarine maneuvering motions is demonstrated.

1. Introduction

The maneuverability of submarines is crucial from the perspective of mission performance. Unlike surface piercing vessels, submarines operate in a state of neutral buoyancy, placing greater emphasis on vertical movement. In particular, manoeuvres such as depth control and emergency rising may be required, necessitating actions not typically considered for surface piercing vessels. Moreover, manoeuvring motions such as depth control and emergency rising can have large drift angles and angles of attack. Therefore, during the initial design stage of submarines, it is essential to carefully consider their maneuverability, giving more rigorous attention to six degrees of freedom (6DOF) motion and ensuring thorough consideration of large drift and/or angle of attack motion. Methods for estimating the maneuverability of submarines during the design stage include numerical simulations based on mathematical models and free-running model tests, similar to those used for surface vessels. To perform numerical simulations based on mathematical models, it is necessary to obtain manoeuvring coefficients of

hydrodynamic forces and moments required in the mathematical models. These manoeuvring coefficients can be obtained through captive model tests, computational fluid dynamics (CFD), and empirical formulas. Furthermore, free-running model tests simulate manoeuvring motions identical to those of full-scale maneuvering motions, using model tests or CFD.

Recently, thanks to the significant advances in high performance computing (HPC), CFD is frequently utilized for obtaining manoeuvring coefficients of hydrodynamic forces and moments and conducting free-running analyses. If there are no constraints in terms of time, cost, and facilities, model tests are preferred. In most cases, mathematical model-based numerical simulations are primarily used to estimate the maneuvering performance of submarines. Representative methods for submarine maneuvering motions, such as those proposed by Gertler and Hagen (1976), Feldman (1979, 1987), and Kim et al. (2021), utilize a six degrees of freedom (6DOF) manoeuvring mathematical model. These methods assume forces and moments acting on the submarine hull are composed of multivariable functions of motion variables and are expressed using Taylor 3rd order approximation.

Notably, the manoeuvring mathematical models for submarines

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Nomenclature

A_{ji}	Added mass or added moment of inertia matrix	[kg]	T	Thrust of propeller	[N]
C_L, C_D	Lift and drag coefficient non-dimensionalized by $\frac{1}{2}\rho L^3 V^2$	[kg•m ²]	t	Propeller deduction factor	[-]
C_{D90}	Cross-flow drag coefficient non-dimensionalized by $\frac{1}{2}\rho L^3 V^2$	[-]	V, U_i	Speed of submarine, component speed at body axis	[m/s]
C'_{LASF}, C'_{LASA}	Cross-flow lift coefficient at bow/stern non-dimensionalized by $\frac{1}{2}\rho L^3 V^2$	[-]	u, v, w	Component speed at body axis	[m/s]
C'_{LAS}	Summed cross-flow lift coefficient	[-]	$\dot{u}, \dot{v}, \dot{w}$	Component acceleration at body axis	[m/s ²]
C'_F	Friction coefficient non-dimensionalized by $\frac{1}{2}\rho L^3 V^2$	[-]	X, Y, Z	Forces at body axis	[N]
$\Delta C'_L, \Delta C'_D$	Coefficient of stall	[-]	X', Y', Z'	Non-dimensional forces at body axis ($= \frac{1}{2}\rho L^2 V^2$)	[-]
D_{ST}'	Correction coefficient by stall	[-]	X_{LAS}, X_F	Axial forces for cross-flow lift and friction	[N]
$[E]$	Euler angle transformation matrix	[-]	x_G, y_G, z_G	Coordinate of center of gravity in body axis	[m]
$\vec{F}, \vec{F}_j$	Force vectors ($j = 1 \sim 3$)	[N]	x'_l, x'_t	Non-dimensional assumed moment levers from midship	[-]
I	Moment of inertia	[kg•m ²]	Y_{Lv}, Y_{Di}, Y_C	Non-dimensional lateral force component due to lift, drag, and cross-flow drag	[-]
I_{xx}, I_{yy}, I_{zz}	Principal moment of inertia at axis	[kg•m ²]	Z_{Lw}, Z_{Di}, Z_C	Non-dimensional vertical force component due to lift, drag, and cross-flow drag	[-]
K, M, N	Moments at axis at body axis	[N•m]	Greek letters		
K', M', N'	Non-dimensional moment at body axis ($= \frac{1}{2}\rho L^3 V^2$)	[-]	α	Angle of attack	[deg.]
L, L_{pp}	Overall length of submarine	[m]	β	Drift angle	[deg.]
L_v, D_i, L_w, L_{ST}	Viscous lift at horizontal plane, induced drag, viscous lift at vertical plane, correction force for stall	[N]	$\delta_r, \delta_s, \delta_b$	Vertical, horizontal rudder deflection at stern and fin deflection	[rad]
$\vec{M}, \vec{M}_j$	Moment vectors ($j = 1 \sim 3$)	[N•m]	ϵ_{jkl}	Sign convertor	[-]
l_{v0}, l_{d0}	Moment levers for lift and drag non-dimensionalized L	[-]	Ω_i	Angular velocity	[rad/s]
M'_{Lv}, M'_{Di}, M'_C	Pitch moments by L_w, D_i , and cross-flow drag, pitch munk moment	[N•m]	ρ	Fluid density	[kg/m ³]
M'_{Munk}					
m	Mass of submarine	[kg]	θ, ϕ, ψ	Pitch, Roll, and Yaw angle	[rad]
m'_x, m'_y, m'_z	Non-dimensionalized added-mass at axis	[-]	Sub-/Super script		
m'_0, m'_1	Non-dimensional moments of lateral projected area of ship about midship	[-]	Gr, B, I, D, C	Gravity, Buoyancy, Inertia, Damping, Control	[-]
N'_{Lv}, N'_{Di}, N'_C	Yaw moments by L_w, D_i , and cross-flow drag, yaw munk moment	[-]	v, h	Vertical, Horizontal	[-]
N'_{Munk}					
p, q, r	Angular velocity at body axis				
$\dot{p}, \dot{q}, \dot{r}$	Angular acceleration at body axis	[rad/m ²]			
p_{corr}	Correction factor for cross-flow drag	[-]			

become more complicated compared to those of surface piercing vessels due to the formers' 6 DOF motion and some coupling effects that occur from their 6DOF motion. These mathematical models are formulated considering both accuracy and complexity of equations, based on the assumption that the submarine motion is not excessive. However, when submarines engage in low-speed manoeuvres or emergency manoeuvres, the assumption of motion not being excessive becomes invalid, thus making the Taylor 3rd order approximation unreliable.

To address this drawback, extending Taylor approximation to higher-order terms results in a more complex expression of hydrodynamic forces and moments. This situation also arises in manoeuvring motions in harbours for surface piercing vessels. In order to overcome this drawback, research has been conducted to simplify the representation of hydrodynamic forces and moments while more accurately reflecting physical phenomena. Hooft (1994) proposed a mathematical model based on slender body theory and cross-flow drag. Bohlmann (1990) applied a mathematical model similar to Hooft's proposal to the 6-DOF maneuvering motion of submarines. Shin (2007) applied a similar mathematical model to submarines as Bohlmann's and established the validity of the proposed mathematical model using theoretical formulas for various submarines. Additionally, Watt et al. (2007) proposed a mathematical model based on a similar form to that suggested by Gertler and Hagen (1967), incorporating a Fourier series-type multivariable function for hydrodynamic forces and moments acting

on submarines manoeuvring with various combinations of both drift angles and angle of attacks. Park et al. (2017) proposed a similar mathematical model to that of Watt (2007) using a polynomial multivariable function instead of the Fourier series-type multivariable function for hydrodynamic forces and moments.

Meanwhile, the mathematical model proposed by Karasuno et al. (1991, 1992) characterizes the forces and moments acting on fishing vessels manoeuvring with drift motions and turning motions using lift, drag, and cross-flow drag coefficients. Although fundamentally similar to Hooft's proposed mathematical model, the model by Karasuno et al. (1991, 1992) has a different form. Karasuno's mathematical model has notable advantages: it may not require model experiments for identifying the hydrodynamic forces related to the rotational motion and it can provide more accuracy in both large drift angles and/or large angles of attack motion. Generally, there are many experimental and/or calculation matrices to identify the manoeuvring hydrodynamic forces when evaluating the manoeuvrability of the vessels. In addition, when the manoeuvrability in large drift (or angle of attack) motion should be evaluated, additional terms in the Taylor expansion based mathematical model should be considered and additional experimental (or calculation) matrices for these terms should be added. In this kind of situation, the advantages of Karasuno's model are that it can help reduce the model experiment matrices and it keeps the original mathematical model without modifying its original form. Despite the advantages of

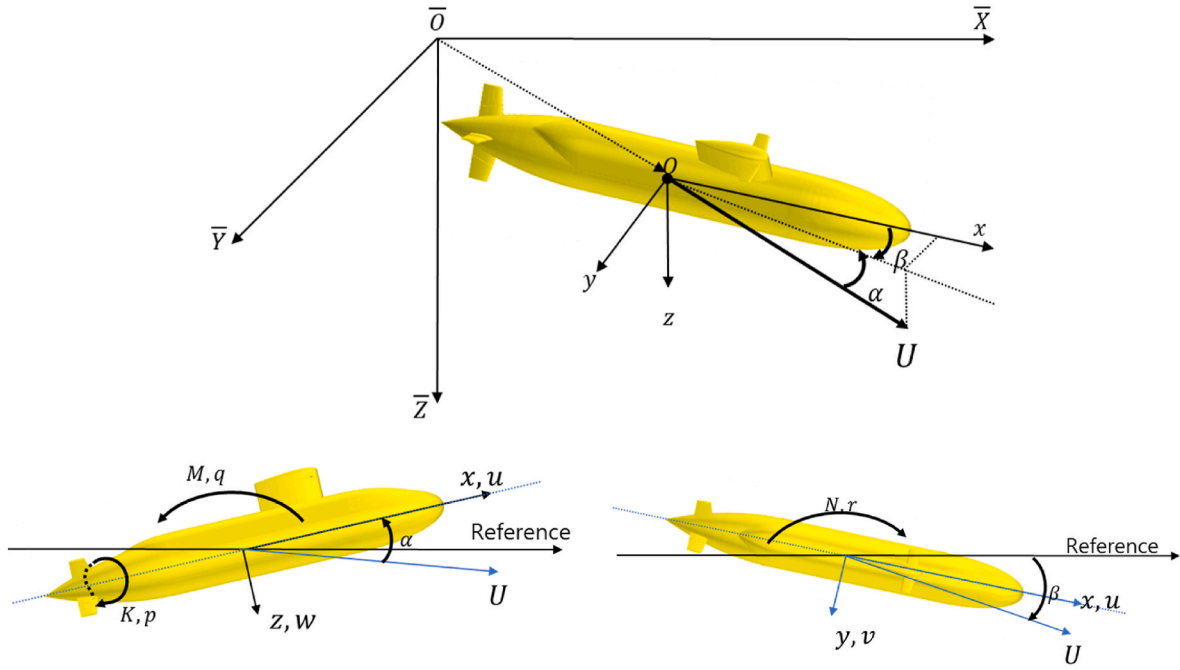


Fig. 1. Coordinate systems.

Karasuno's model, it has not been used for the equation of manoeuvring motion for other vessels except fishing vessels. Therefore, this study is focused on the applicability of Karasuno's model to the equations of manoeuvring motion for the submarine.

Keeping in mind these advantages of Karasuno's model, it can be considered valuable for not only describing maneuvering motions, including emergency manoeuvring motions such as emergency rising, but also the normal manoeuvring motions of submarines. After formulating a new type of mathematical model from the original Karasuno model, this study discusses only normal manoeuvring motions such as zig-zag motions in both the horizontal and vertical planes and turning motions. The reason for this limited focus is that the present study encompasses the initial stage of applying a new type of mathematical model for submarines and is focused on validating the applicability of this new mathematical model.

The purpose of this study thus is to validate the applicability of the proposed mathematical model for identifying the hydrodynamic coefficients acting on the submarine's hull and to confirm that this model can be useful to predict the submarine's normal manoeuvring motion such as zig-zag motion and turning in both the horizontal and vertical planes. The findings contribute to identifying the manoeuvring hydrodynamic forces for submarines and evaluating their manoeuvring performance.

The application of the Karasuno model to the forces and moments acting on submarines' hulls maneuvering with horizontal drift angles and vertical angle of attacks was previously reviewed by the authors (Choi et al., 2023). The present study builds on the research conducted by Choi et al. (2023), with the aim of evaluating the applicability of the Karasuno model to analyze the forces and moments acting on submarines' hulls during turning motions combined with drift motions. Chapter 2 introduces the equations for the 6DOF manoeuvring motion of submarines and the fluid dynamic mathematical model proposed by Karasuno et al. In chapter 3, validation studies for a CFD analysis are performed to confirm whether the results from the CFD in this study are useful. In particular, the drift and rotating cases for the DARPA Suboff submarine are selected because both CFD results and experiment results are available. In chapter 4, various CFD calculations including both horizontal and vertical motions for the MARIN BB2 submarine are performed to obtain the hydrodynamic coefficients for the Karasuno

model composing the equations of 6DOF manoeuvring motion. In addition, the detailed procedure to obtain the manoeuvring coefficient for the Karasuno model is introduced and the hydrodynamic coefficients proposed by Karasuno et al. (1991, 1992) are derived from these calculations. In chapter 5, calculations for vertical and horizontal zig-zag motions are performed and compared to the free-running model tests conducted by Overpelt et al. (2015) to validate the mathematical model.

2. Physical based mathematical model

In this section, we discuss the physical based mathematical model for the manoeuvring motion of a submarine.

2.1. Equation of manoeuvring motion for submarine

Assuming the submarine as a rigid body and symmetric about the longitudinal axis, a body-fixed coordinate system O-xyz fixed to the submarine is established as shown in Fig. 1. From Newton's equations of motion, the equations of 6DOF maneuvering motion for the forces and moments of the submarine in each component are expressed as Eq. (1) as follows (Shon et al., 2006):

$$\begin{aligned}
 X &= m\{\dot{u} + qw - ru - x_G(q^2 + r^2) + z_G\} \\
 Y &= m\{\dot{v} + ru - pw + z_G(qr - \dot{p}) + x_G(q\dot{p} + \dot{r})\} \\
 Z &= m\{\dot{w} + pv - qu - z_G(p^2 + q^2) + x_G(rp - \dot{q})\} \\
 K &= I_{xx}\dot{p} + (I_{zz} - I_{yy})qr - m\{z_G(\dot{v} + ru - pw) + x_Gz_G(pq + r^2)\} \\
 M &= I_{yy}\dot{q}(I_{xx} - I_{zz})rp \\
 &\quad + m\{z_G(\dot{u} + qw - ru) - x_G(\dot{w} + pv - qu) + x_Gz_G(p^2 - r^2)\} \\
 N &= I_{zz}\dot{r}(I_{yy} - I_{xx})pq + m\{x_G(\dot{v} + ru - pw) + z_Gx_G(rq - p^2)\}
 \end{aligned} \quad (1)$$

The forces and moments on the left-hand side of Eq. (1) can be expressed in vector form and represented as the sum of individual forces and moments, as shown in Eq. (2), divided into their respective components.

$$\begin{aligned}
 \vec{F} &= \vec{F}_{Gr} + \vec{F}_B + \vec{F}_I + \vec{F}_D + \vec{F}_C \\
 \vec{M} &= \vec{M}_{Gr} + \vec{M}_B + \vec{M}_I + \vec{M}_D + \vec{M}_C
 \end{aligned} \quad (2)$$

1 Gravity force

$$\begin{aligned}\vec{F}_{Gr} &= [E]mg\vec{k} \\ \vec{M}_{Gr} &= \vec{r}_G \times ([E]mg\vec{k})\end{aligned}\quad (3)$$

Here, $\vec{r}_G$ is the positional vector for the center of gravity of the submarine and $\vec{k}$ is the unit vector of the $\bar{Z}$ axis.

$[E]$ is the transformation matrix that converts the space-fixed coordinate system to the body-fixed coordinate system, defined as follows (Abkowitz, 1964). However, ϕ , θ , ψ represent the Euler angles.

$$[E] = \begin{bmatrix} \cos \theta \cos \psi & \cos \theta \sin \psi & -\sin \theta \\ \sin \phi \sin \theta \cos \psi & \sin \phi \sin \theta \sin \psi & \sin \phi \cos \theta \\ -\cos \phi \sin \psi & +\cos \phi \cos \psi & \\ \cos \phi \sin \theta \cos \psi & \cos \phi \sin \theta \sin \psi & \sin \phi \cos \theta \\ +\sin \phi \sin \psi & -\sin \phi \cos \psi & \end{bmatrix}\quad (4)$$

2 Buoyancy force

$$\begin{aligned}\vec{F}_B &= -[E]\rho g \nabla \vec{k} \\ \vec{M}_B &= -\vec{r}_B \times ([E]\rho g \nabla \vec{k})\end{aligned}\quad (5)$$

Here, $\vec{r}_B$ is the positional vector for the center of gravity of the submarine

3 Inertia force

In an inviscid infinite fluid, the inertial force or added-mass force acting on an object in 6DOF motion is expressed as follows, according to Newman (1978).

$$\begin{aligned}F_j &= -\dot{U}_i A_{ji} - \epsilon_{jkl} U_i \Omega_k A_{li} \\ M_j &= -\dot{U}_i A_{j+3,i} - \epsilon_{jkl} U_i \Omega_k A_{l+3,i} - \epsilon_{jkl} U_i U_k A_{li}\end{aligned}\quad (6)$$

A_{ji} denotes the added-mass or added-inertia moment in the j -direction resulting from the i -direction motion mode, referred to as the added-mass tensor. ϵ_{jkl} is a symbol, taking a positive (+) sign for permutations 123, 231, and 312, a negative (−) sign for permutations 132, 321, and 213, and zero (0) for any other permutations. The meaning of each sign is as follows:

$i: 1 \sim 6, j, k, l: 1 \sim 3$

$$\begin{aligned}U_i &: (U_1, U_2, U_3) \equiv (u, v, w), \\ &(U_4, U_5, U_6) \equiv (p, q, r) \\ \Omega_i &: (\Omega_1, \Omega_2, \Omega_3) \equiv (p, q, r)\end{aligned}$$

Referring to the fundamental equation of 6DOF motion represented by Eq. (1), one can utilize Eq. (4) to specifically calculate each component of the inertial hydrodynamic force.

4 Viscous damping force

The hydrodynamic forces due to fluid viscosity are determined through model experiments. The viscous damping hydrodynamic forces are determined by constructing a mathematical model in the form of a Taylor series function, assuming these forces to be polynomial functions of motion variables related to submarine motion. By applying the mathematical model to the forces and moments obtained through model experiments, the coefficients constituting the mathematical model are determined. This approach is used to determine the viscous damping hydrodynamic forces and moments. As mentioned in the introduction, these forces and moments can be expressed by the equations proposed by Gertler and Hagen (1967) or Kim et al. (2021). As also already mentioned, the purpose of this study is to propose a new mathematical

model for the manoeuvring motion of submarines applying a physical based approach proposed by Karasuno to these external forces and to verify both the validity and applicability of the proposed mathematical model for the manoeuvring motion of submarines. Equation (7) represents the viscous damping hydrodynamic forces and moments acting on the submarine's hull, as proposed by Kim et al. (2021).

$$\begin{aligned}X_D &= X_{\dot{u}}\dot{u} + X_{vr}vr + X_{wq}wq + X_{qq}q^2 + X_{rr}r^2 + X_{pr}pr \\ &\quad + X_{vv}v^2 + X_{ww}w^2 + X_{uu}u^2 \\ Y_D &= Y_0 + Y_{\dot{v}}\dot{v} + Y_{\dot{r}}\dot{r} + Y_{\dot{p}}\dot{p} + Y_{pq}pq + Y_{qr}qr + Y_{wp}wp \\ &\quad + Y_{vr}vr + Y_{r|r}r|r| + Y_{vv}v + Y_{v|v}|v| + Y_{vvr}v^2r + Y_{vrr}vr^2 \\ Z_D &= Z_0 + Z_{\dot{w}}\dot{w} + Z_{\dot{q}}\dot{q} + Z_{pp}p^2 + Z_{pr}pr + Z_{qq}q + Z_{q|q}|q| \\ &\quad + Z_{vp}vp + Z_{wv}w + Z_{w|w}|w| + Z_{vv}v^2 + Z_{ww}w^2 + Z_{wqq}wq^2 \\ K_D &= K_0 + K_{\dot{v}}\dot{v} + K_{\dot{p}}\dot{p} + K_{\dot{r}}\dot{r} + K_{qr}qr + K_{pq}pq + K_{vq}vq \\ &\quad + K_{wp}wp + K_{wr}wr + K_{pp}p + K_{rr}r + K_{vv}v + K_{v|v}|v| + K_{vvw}vw \\ M_D &= M_0 + M_{\dot{w}}\dot{w} + M_{\dot{q}}\dot{q} + M_{pp}p^2 + M_{rr}r^2 + M_{pr}pr + M_{vp}vp \\ &\quad + M_{vr}vr + M_{qq}q + M_{q|q}|q| + M_{ww}w + M_{w|w}|w| + M_{vv}v^2 \\ &\quad + M_{wv}wq + M_{wqq}wq^2 \\ N_D &= N_0 + N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} + N_{\dot{p}}\dot{p} + N_{pq}pq + N_{qr}qr + N_{vq}vq \\ &\quad + N_{pp}p + N_{wp}wp + N_{rr}r + N_{r|r}r|r| + N_{vv}v + N_{v|v}|v| \\ &\quad + N_{vvr}v^2r + N_{vrr}vr^2\end{aligned}\quad (7)$$

Here, the force and moment in Eq. (7) are non-dimensionalized as follows:

$$Force' = \frac{Force}{0.5\rho L^2 V^2}, \quad Moment' = \frac{Moment}{0.5\rho L^3 V^2}$$

5 Control forces

The hydrodynamic forces and moments generated by the control surfaces and propulsors installed on submarines are conventionally denoted in the field of maneuverability as follows:

$$\begin{aligned}F_C &= \begin{bmatrix} X_{\delta_r\delta_r}\delta_r^2 + X_{\delta_s\delta_s}\delta_s^2 + X_{\delta_b\delta_b}\delta_b^2 + (1-t)T \\ Y_{\delta_r}\delta_r \\ Z_{\delta_s}\delta_s + Z_{\delta_b}\delta_b \end{bmatrix} \\ M_C &= \begin{bmatrix} K_{\delta_r}\delta_r \\ M_{\delta_s}\delta_s + M_{\delta_b}\delta_b \\ N_{\delta_r}\delta_r \end{bmatrix}\end{aligned}\quad (8)$$

Here, δ_r represents the rudder deflection, δ_s the stern plane deflection, δ_b the fin deflection, t the thrust deduction ratio, and T the propulsive thrust.

2.2. Physical based mathematical model for forces and moments

2.2.1. Elemental component of hydrodynamic forces and moments

Karasuno et al. (1991, 1992) expanded their mathematical model that was originally developed for fishing vessels navigating with drift motion at low speeds to include hydrodynamic forces and moments during turning motion. Furthermore, Choi et al. (2023) validated the applicability of Karasuno's model, initially proposed for fishing vessels in drift motion, when applied to submarine hulls. This study examines an extension of Karasuno's mathematical model for hydrodynamic forces and moments acting on vessels maneuvering with horizontal and/or vertical turning motion with combined drift angle for application to submarines in 6DOF manoeuvring motion. To accomplish this, it is necessary to examine the mathematical model for hydrodynamic forces and moments acting on a vessel maneuvering with horizontal-plane

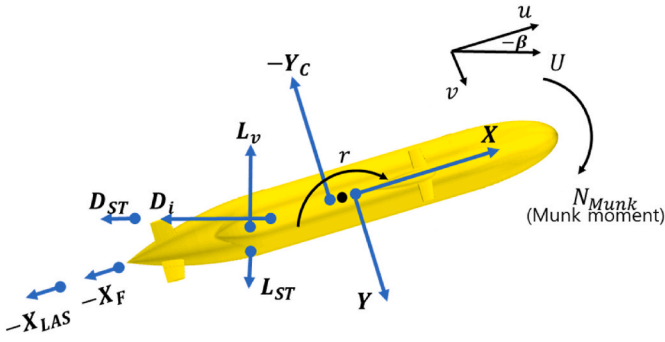


Fig. 2. Elemental components of hydrodynamic force acting on the submarine (horizontal motion).

Table 1

Hydrodynamic force component acting on the hull with drift motion.

Component force and/or moment	Mathematical expression
1 Munk moment	$(m'_x - m'_y)u'v'$
2 Lift	$L_v = \frac{1}{2}\rho L_{pp} dU^2 C_L' u'v'$
3 Drag	$D_i = \frac{1}{2}\rho L_{pp} dU^2 C_D' u'^2 v'^2$
4 Cross-flow drag	$Y_c = \frac{1}{2}\rho L_{pp} dU^2 \int C_{D90} v'^3 (1 + p \cdot u'^2) d' \cdot dx' = \frac{1}{2}\rho L_{pp} dU^2 C_{D90} m_0' v'^3$ $N_c = \frac{1}{2}\rho L_{pp}^2 dU^2 \int C_{D90} v'^3 (1 + p \cdot u'^2) x' d' \cdot dx' = \frac{1}{2}\rho L_{pp}^2 dU^2 C_{D90} m_1' v'^3$
5 Cross-flow lift	$X_{LAS} = \frac{1}{2}\rho L_{pp} dU^2 \cdot \{C'_{LASF} - C'_{LASA}\} v'^2 = \frac{1}{2}\rho L_{pp} dU^2 C'_{LAS} v'^2$
6 Friction	$X_F = \frac{1}{2}\rho L_{pp} dU^2 C'_F u' u'$
7 Correction force for stall effect	$(L'_{ST} + D'_{ST})_x = -\Delta C'_L u'v'^2 + \Delta C'_{D_i} u'^3 v'^2$

turning motion, as proposed by Karasuno et al. (1991, 1992). Fig. 2 depicts the forces and moments acting on a submarine in horizontal turning motion, separated into hydrodynamic force components proposed by Karasuno et al. (1991, 1992). As shown in Fig. 2, Karasuno et al. decomposed the hydrodynamic forces and moments acting on the hull into ideal fluid forces (N_{Munk}), viscous lift (L_v), induced drag (D_i), cross-flow drag (Y_c), cross-flow lift (X_{LAS}), hull friction drag (X_F), and the stall effect ($L_{ST} + D_{ST}$). The ideal fluid force component refers to the forces and moments due to added mass related to the vessel's motion direction, while the viscous lift and induced drag refer to lift and drag addressed in wing theory. These two components resemble the linear and nonlinear terms in the manoeuvring hydrodynamic coefficients using Taylor 3rd order approximation but can consider more nonlinear effects. Cross-flow drag and cross-flow lift denote the drag and lift generated as the vessel's drift angle increases, respectively. While Hoofft (1994) only considered cross-flow drag, Karasuno et al. took into account cross-flow lift to consider the increase in longitudinal force due to the increase in drift angle. Additionally, hull friction drag signifies the force generated by fluid viscosity. Lastly, the stall effect is considered to correct the longitudinal force acting on the ship in large drift motion. Table 1 presents the hydrodynamic force components proposed by Karasuno et al. for drift motion.

The hydrodynamic force components (lift, induced drag, cross-flow lift, cross-flow drag, etc.) acting on the hull depicted in Fig. 2 play a crucial role not only in representing the forces and moments acting on a ship in drift motion but also on a ship in turning motion. Karasuno et al. proposed that coefficients constituting each hydrodynamic force

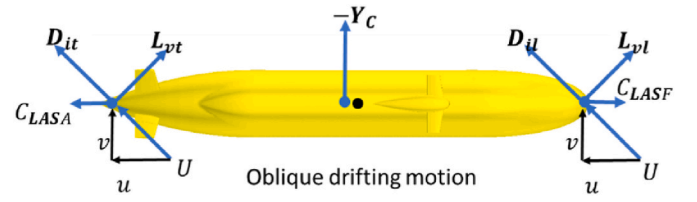


Fig. 3. Elemental components of hydrodynamic force at oblique motion (horizontal).

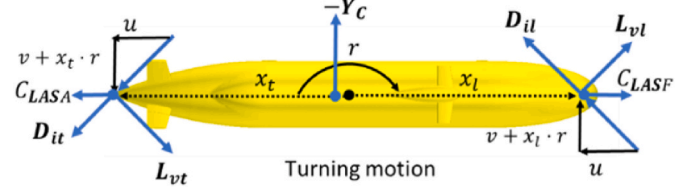


Fig. 4. Elemental components of hydrodynamic force at turning motion (horizontal).

component could be obtained from the forces and moments acting on the ship in drift motion, offering an advantage in expressing the forces and moments on the ship in turning motion. This stands in contrast to traditional mathematical models based on Taylor series expansions, where obtaining hydrodynamic force coefficients (e.g., Y_r , Y_{rr} , Y_{vr} , Y_{vrr} , N_r , N_{rr} , N_v , N_{vr} , N_{vrr} , etc.) acting on the ship in turning motion requires conducting rotating arm tests (or pure yaw tests and combined yaw tests in PMM test). As mentioned before, the hydrodynamic forces and moments related to the rotational motion can be obtained from the drift motion (refer to Eq. (10) and Eq. (20)).

To apply the mathematical model proposed by Karasuno et al. for 3DOF manoeuvring motion to the 6DOF manoeuvring motion of a submarine, it is necessary to separate the submarine's motion into horizontal and vertical components. Therefore, in the following sections, each of these motions are described separately.

2.2.2. Hydrodynamic forces and moments for horizontal manoeuvring motion

Karasuno et al. assumed that among the components of hydrodynamic forces acting on a ship in horizontal motion, lift, induced drag, and cross-flow lift components are separated and act independently at the bow and stern. These three components are treated as presented in Figs. 3 and 4 for the drift motion and the turning motion, respectively.

First, considering only the drift motion of the hull, as presented in Fig. 3, it is assumed that there are no components dependent on the rotational velocity ($v \neq 0, r = 0$), and the lift, drag, and cross-flow lift acting on the hull are assumed to be separated at the bow and stern ends, respectively. Therefore, the lift, drag, and cross-flow lift acting at the bow and stern ends can be expressed as follows in Eq. (9)–(11).

$$\begin{aligned} X' &= (C'_L - \Delta C'_L) u' v'^2 - (C'_{D_i} - \Delta C'_{D_i}) u'^3 v'^2 - C'_F |u'| u' - C'_{LAS} v'^2 \\ &= C'_L u' v'^2 - C'_{D_i} u'^3 v'^2 - X'_F - X'_{LAS} - (L'_{ST} - D'_{ST})_x \\ &= X'_{L_v} - X'_{D_i} - X'_F - X'_{LAS} - (L'_{ST} - D'_{ST})_x \end{aligned} \quad (9)$$

$$\begin{aligned} Y' &= -(C'_L + C'_{L_t}) u'^2 v' - (C'_{D_{il}} + C'_{D_{il}}) u'^2 v'^3 \\ &\quad - \int C'_{D90} v'^3 (1 + p \cdot u'^2) d' \cdot dx' \\ &= -C'_L u'^2 v' - C'_{D_i} u'^2 v'^3 - \int C'_{D90} v'^3 (1 + p \cdot u'^2) d' \cdot dx' \\ &= -L'_{L_v} - D'_{D_i} - Y'_c = -Y'_{L_v} - Y'_{D_i} - Y'_c \end{aligned} \quad (10)$$

$$\begin{aligned}
N' &= (m_x' - m_y') u' \dot{v}' - (C_{L_t}' x_t' + C_{L_t}' x_t') u'^2 \dot{v}' \\
&\quad - (C_{D_{il}}' x_t' + C_{D_{il}}' x_t') u'^2 \dot{v}'^3 - \int C_{D_{90}}' v'^3 (1 + p \cdot u'^2) x' d' \cdot dx' \\
&= (m_x' - m_y') u' \dot{v}' - x_t' (C_{L_t}' - C_{L_t}') u'^2 \dot{v}' - x_t' (C_{D_{il}}' - C_{D_{il}}') u'^2 \dot{v}'^3 \\
&\quad - \int C_{D_{90}}' v'^3 (1 + p \cdot u'^2) x' d' \cdot dx' \\
&= (m_x' - m_y') u' \dot{v}' - C_{L_t}' L_{v0} u'^2 \dot{v}' - C_{D_{il}}' L_{v0} u'^2 \dot{v}'^3 \\
&\quad - \int C_{D_{90}}' v'^3 (1 + p \cdot u'^2) x' d' \cdot dx' \\
&= M_{MUNK}' - Y_{L_v}' \cdot L_{v0}' - Y_{D_{il}}' \cdot L_{v0}' - M_c' = M_{MUNK}' - N_{L_v}' - N_{D_{il}}' - M_c'
\end{aligned} \quad (11)$$

Furthermore, in the case of pure rotating motion ($v = 0, r \neq 0$), the lateral velocity at the bow and stern can be expressed as the sum ($v + x_t \cdot r$ and $v + x_l \cdot r$) of the pure lateral velocity (v) and the tangential velocity due to pure rotating motion ($x_t \cdot r$ and $x_l \cdot r$), as shown in Fig. 4. It is assumed that this velocity at the bow and stern acts as the cause of lift, drag, and cross-flow lift generation. Therefore, in the case of pure rotational motion only ($v = 0, r \neq 0$), the forces and the moment due to the lift, drag, and cross-flow lift generated at the bow and stern can be expressed as shown in Eq. (12)–(14).

$$\begin{aligned}
X' &= \left\{ (C_{L_t}' - \Delta C_{L_t}^h) \cdot (x_t' r')^2 + (C_{L_t}' - \Delta C_{L_t}^h) \cdot (x_t' r')^2 \right\} u' \\
&\quad - \left\{ (C_{D_{il}}' - \Delta C_{D_{il}}^h) \cdot (x_t' r')^2 + (C_{D_{il}}' - \Delta C_{D_{il}}^h) \cdot (x_t' r')^2 \right\} u'^3 \\
&\quad - C_F^h |u| u' \\
&\quad - C_{LAS}^h \frac{1}{4} r'^2 \left\{ (C_{L_t}' - \Delta C_{L_t}^h) + (C_{L_t}' - \Delta C_{L_t}^h) \right\} u' (x_t' r')^2 \\
&\quad - \left\{ (C_{D_{il}}' - \Delta C_{D_{il}}^h) + (C_{D_{il}}' - \Delta C_{D_{il}}^h) \right\} u'^3 (x_t' r')^2 - C_F^h |u| u' \\
&\quad - C_{LAS}^h \frac{1}{4} r'^2 \\
&= (C_{L_t}' - \Delta C_{L_t}^h) u' x_t'^2 - (C_{D_{il}}' - \Delta C_{D_{il}}^h) u'^3 (x_t' r')^2 - C_F^h |u| u' \\
&\quad - C_{LAS}^h \frac{1}{4} r'^2
\end{aligned} \quad (12)$$

$$\begin{aligned}
Y' &= -m_x' u' r' - (C_{L_t}' x_t' + C_{L_t}' x_t') u'^2 r' - (C_{D_{il}}' x_t' - C_{D_{il}}' x_t') u'^2 x_t' r'^3 \\
&\quad - \int C_{D_{90}}' x' r'^3 (1 + p^h \cdot u'^2) d' \cdot dx' \\
&= -m_x' u' r' - (C_{L_t}' x_t' + C_{L_t}' x_t') u'^2 r' - (C_{D_{il}}' x_t' - C_{D_{il}}' x_t') u'^2 x_t' r'^3 \\
&\quad - \int C_{D_{90}}' x' r'^3 (1 + p^h \cdot u'^2) d' \cdot dx' \\
&= -m_x' u' r' - (C_{L_t}' - C_{L_t}') u'^2 x_t' r' - (C_{D_{il}}' - C_{D_{il}}') u'^2 (x_t' r')^3 - Y_{c-rotating}' \\
&= -m_x' u' r' - C_{L_t}' C_{v0}^h u'^2 x_t' r' - C_{D_{il}}' C_{i0}^h u'^2 (x_t' r')^3 - Y_{c-rotating}'
\end{aligned} \quad (13)$$

$$\begin{aligned}
N' &= -x_t' (C_{L_t}' + C_{L_t}') u'^2 (x_t' r') - x_t' (C_{D_{il}}' + C_{D_{il}}') u'^2 (x_t' r')^3 \\
&\quad - \int C_{D_{90}}' (x' r')^3 (1 + p^h \cdot u'^2) x' d' \cdot dx' \\
&= -x_t' C_{L_t}' u'^2 (x_t' r') - x_t' C_{D_{il}}' u'^2 (x_t' r')^3 \\
&\quad - \int C_{D_{90}}' (x' r')^3 (1 + p^h \cdot u'^2) x' d' \cdot dx'
\end{aligned} \quad (14)$$

Consequently, a turning motion ($v \neq 0, r \neq 0$) can be considered as a combined motion of drift and rotation. Therefore, the forces and moments acting on the ship during turning can be represented by combining Eq. (9)–(11) and Eq. (12)–(14) above, as shown in Eq. (15)–(17).

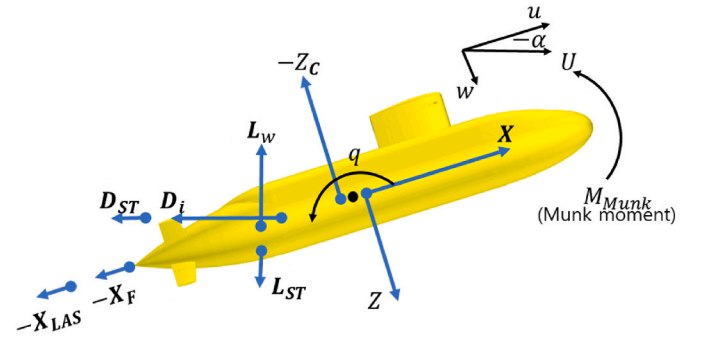


Fig. 5. Elemental components of hydrodynamic force acting on the submarine (vertical motion).

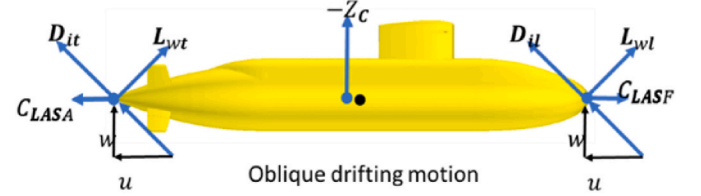


Fig. 6. Elemental components of hydrodynamic force at oblique motion (vertical).

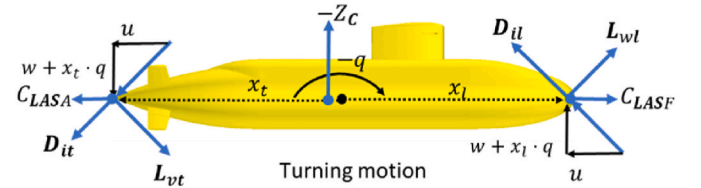


Fig. 7. Elemental components of hydrodynamic force at turning motion (vertical).

$$\begin{aligned}
X' &= m_y' \dot{v}' r' + (C_{L_t}' - \Delta C_{L_t}^h) u' (v' + x_t' r')^2 \\
&\quad - (C_{D_{il}}' - \Delta C_{D_{il}}^h) u'^3 (v' + x_t' r')^2 - C_F^h |u| u' \\
&\quad - C_{LAS}^h \left(v' + \frac{1}{4} r'^2 \right) + C_{LASAF}^h v' r'
\end{aligned} \quad (15)$$

$$\begin{aligned}
Y' &= -m_x' u' r' - C_{L_t}' u'^2 (v' + C_{v0}^h x_t' r') - C_{D_{il}}' u'^2 (v' + C_{i0}^h x_t' r')^3 \\
&\quad - \int C_{D_{90}}' (v' + x' r')^3 (1 + p^h \cdot u'^2) d' \cdot dx'
\end{aligned} \quad (16)$$

$$\begin{aligned}
N' &= (m_x' - m_y') u' \dot{v}' - x_t' C_{L_t}' u'^2 (C_{v0}^h v' + x_t' r') \\
&\quad - x_t' C_{D_{il}}' u'^2 (C_{i0}^h v' + x_t' r')^3 \\
&\quad - \int C_{D_{90}}' (v' + x' r')^3 (1 + p^h \cdot u'^2) x' d' \cdot dx'
\end{aligned} \quad (17)$$

2.2.3. Hydrodynamic forces and moments for vertical manoeuvring motion

In the previous section, the hydrodynamic forces acting on the ship during horizontal turning motion were examined. The same approach as presented earlier can be applied to vertical turning motion as well. If we represent the hydrodynamic force components acting on the ship during vertical turning motion, as shown in Figs. 5–7, then the resultant forces and moments acting on the ship can be expressed as Eq. (18)–(20).

$$\begin{aligned} X' &= m_x' \dot{w}' \dot{q}' + (C_L^{v'} - \Delta C_L^{v'}) u' (w' + x_i' \dot{q}')^2 \\ &\quad - (C_{D_i}^{v'} - \Delta C_{D_i}^{v'}) u'^3 (w' + x_i' \dot{q}')^2 \\ &\quad - C_{LAS}^{v'} \left(w' + \frac{1}{4} \dot{q}'^2 \right) + C_{LASAF}^{v'} w' \dot{q}', \end{aligned} \quad (18)$$

$$\begin{aligned} Z' &= -m_x' u' \dot{q}' - C_L^{v'} u'^2 (w' + C_{w0}^{v'} x_i' \dot{q}') - C_{D_i}^{v'} u'^2 (w' + C_{i0}^{v'} x_i' \dot{q}')^3 \\ &\quad - \int C_{D90}^{v'} (w' + x' \dot{q}')^3 (1 + p^v \cdot u^2) d' dx' \end{aligned} \quad (19)$$

$$\begin{aligned} M' &= (m_x' - m_z') u' \dot{w}' - x_i' C_L^{v'} u'^2 (C_{w0}^{v'} w' + x_i' \dot{q}') \\ &\quad - x_i' C_{D_i}^{v'} u'^2 (C_{i0}^{v'} w' + x_i' \dot{q}')^3 \\ &\quad - \int C_{D90}^{v'} (w' + x' \dot{q}')^3 (1 + p^v \cdot u^2) x' d' \cdot dx' \end{aligned} \quad (20)$$

2.2.4. Final hydrodynamic forces and moments for 6DOF manoeuvring motion of submarine

The forces and moments acting on a submarine in 6DOF manoeuvring motion, considering both horizontal and vertical motion as well as the coupling effects, can be summarized as Eq. (21)–(26), as introduced in sections 2.2.2 to 2.2.3. It is assumed that the coupling terms in Eq. (21)–(26) except $Z'_{coupling}$ are small enough to be ignored in this study but this should be confirmed through further investigations. With regard to the $Z'_{coupling}$ term in Eq. (23), the vertical lift and drag component due to the horizontal drift motion were considered in the same manner as for $C_L^{h'}$ and $C_{D_i}^{h'}$ in the Y' force.

In addition, the roll moment can be written as given in Eq. (24). As shown in Eq. (24), the roll moment is derived from the lateral force (Eq. (22)) and the principal moment.

$$\begin{aligned} X' &= m_y' \dot{v}' \\ &\quad + (C_L^{h'} - \Delta C_L^{h'}) u' (\dot{v}' + x_i' \dot{r}')^2 - (C_{D_i}^{h'} - \Delta C_{D_i}^{h'}) u'^3 (\dot{v}' + x_i' \dot{r}')^2 \\ &\quad - C_F^{h'} |u'| \dot{u}' - C_{LAS}^{h'} \left(\dot{v}' + \frac{1}{4} \dot{r}'^2 \right) + C_{LASAF}^{h'} \dot{v}' \dot{r}' + m_z' w' \dot{q}' \\ &\quad + (C_L^{v'} - \Delta C_L^{v'}) u' (w' + x_i' \dot{q}')^2 - (C_{D_i}^{v'} - \Delta C_{D_i}^{v'}) u'^3 (w' + x_i' \dot{q}')^2 \\ &\quad - C_{LAS}^{v'} \left(w' + \frac{1}{4} \dot{q}'^2 \right) + C_{LASAF}^{v'} w' \dot{q}' + X'_{couple} \end{aligned} \quad (21)$$

$$\begin{aligned} Y' &= -m_x' u' \dot{r}' - C_L^{h'} u'^2 (\dot{v}' + C_{v0}^{h'} x_i' \dot{r}') - C_{D_i}^{h'} u'^2 (\dot{v}' + C_{i0}^{h'} x_i' \dot{r}')^3 \\ &\quad - \int C_{D90}^{h'} (\dot{v}' + x' \dot{r}')^3 (1 + p^h \cdot u^2) d' \cdot dx' + Y'_{couple} \end{aligned} \quad (22)$$

$$\begin{aligned} Z' &= -m_x' u' \dot{q}' - C_L^{v'} u'^2 (w' + C_{w0}^{v'} x_i' \dot{q}') - C_{D_i}^{v'} u'^2 (w' + C_{i0}^{v'} x_i' \dot{q}')^3 \\ &\quad - \int C_{D90}^{v'} (w' + x' \dot{q}')^3 (1 + p^v \cdot u^2) d' \cdot dx' + Z'_{couple} \end{aligned} \quad (23)$$

$$K' = L_v' \cdot K_{lv}' + D_i' \cdot K_{di}' + Y_c' \cdot K_c' + K'_{couple} \quad (24)$$

$$\begin{aligned} M' &= (m_x' - m_z') u' \dot{w}' \\ &\quad - x_i' C_L^{h'} u'^2 (C_{w0}^{h'} w' + x_i' \dot{q}') - x_i' C_{D_i}^{h'} u'^2 (C_{i0}^{h'} w' + x_i' \dot{q}')^3 \\ &\quad - \int C_{D90}^{h'} (w' + x' \dot{q}')^3 (1 + p^h \cdot u^2) x' d' \cdot dx' + M'_{couple} \end{aligned} \quad (25)$$

$$\begin{aligned} N' &= (m_x' - m_y') u' \dot{v}' \\ &\quad - x_i' C_L^{h'} u'^2 (C_{v0}^{h'} \dot{v}' + x_i' \dot{r}') - x_i' C_{D_i}^{h'} u'^2 (C_{i0}^{h'} \dot{v}' + x_i' \dot{r}')^3 \\ &\quad - \int C_{D90}^{h'} (\dot{v}' + x' \dot{r}')^3 (1 + p^h \cdot u^2) x' d' \cdot dx' + N'_{couple} \end{aligned} \quad (26)$$

Table 2

Experimental cases for each DARPA Suboff configuration.

	Bare hull	Sail and rudders appended	Model
Configuration index	1	2	3
Resistance	○	○	○
Translation	○	–	–
Turning	○	–	–

Table 3

Principal dimensions of the DARPA Suboff.

Quantity	Symbol	Unit	Model
Overall length	L_{OA}	[m]	4.356
Length between perpendiculars	L_{PP}, l	[m]	4.261
Maximum hull radius	R_{MAX}	[m]	0.254
Maximum hull diameter	d	[m]	0.508
Center of buoyancy	L_{CB}	[m]	0.4621 L_{OA}
Wetted surface area of bare hull	S	[m ²]	5.988
Linear scale ratio	–	–	24

Table 4

Experimental test conditions for Suboff.

Model	$\frac{l}{d}$	$\frac{R}{d}$	$Re (\times 10^6)$	β
SUBOFF (Roddy, 1990)	8.38	–	14.0	[–18°, 18°]
SUBOFF (Etebari et al., 2008)	8.38	2.74 ± 0.04	6.5	[–3.8°, 16.5°]

Table 5

Calculation conditions for CFD calculations.

Subjects	Scheme
URANS solver	Star-CCM + v.16
Turbulence model	RSM(Reynolds Stress Model)
Temporal/Convection term	Backward/2nd order upwind
Grid type	Unstructured trimmer grid
Pressure-velocity coupling	Unsteady SIMPLE

3. Validations of CFD analysis

3.1. Experimental data for validation

Before acquiring the hydrodynamic force components constituting the mathematical model described in Section 2 using computational fluid dynamics (CFD), it is necessary to validate the CFD analysis results. In this section, for the purpose of validation, CFD analyses were conducted and compared with the captive model tests for the DARPA Suboff submarine, as well as with CFD analyses conducted by other institutions (Roddy, 1990; Etebari et al., 2008; Zhang et al., 2013; Toxopeus, 2008; Takahashi and Sahoo, 2019). Roddy (1990) conducted captive model tests by configuring the studied submarine in various combinations, as shown in Table 2, to measure the hydrodynamic forces acting on DARPA Suboff in drift conditions. Etebari et al. (2008) measured the hydrodynamic forces acting on the hull under various drift angles with the submarine in a bare hull condition through captive model tests. In this section, CFD analyses are performed under the same conditions as the model tests conducted by Roddy (1990) and Etebari et al. (2008) for the bare hull of the DARPA Suboff. Tables 3 and 4 respectively present the main specifications and test conditions of the DARPA Suboff used in the model tests conducted by Roddy (1990) and Etebari et al. (2008).

3.2. Numerical setup

The CFD solver used in this study is Siemens STAR-CCM+. The basic

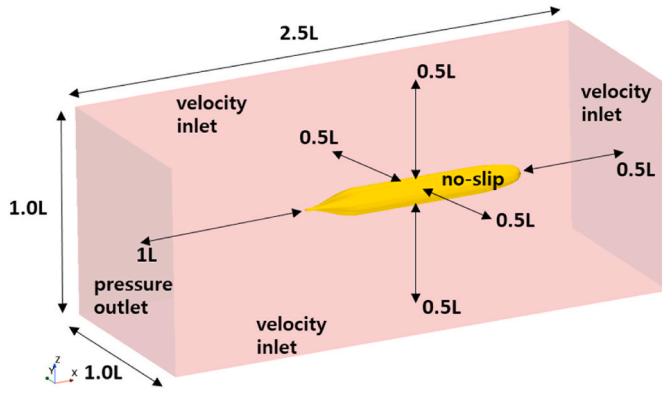


Fig. 8. Computational domain and boundary conditions.

numerical analysis settings applied in the CFD analysis are summarized in Table 5. The CFD analysis was conducted for the submarine's drift motion and turning motion with the combined drift angle. For computational convenience, both types of analysis were performed under conditions where fluid inflow was directed towards the hull instead of simulating the motion of the hull itself. Detailed information regarding this approach is described in the literature (Toxopeus, 2008). In the case of drift motion, the dimensionless speed of drift is described as a function of the drift angle, necessitating careful consideration (Zhang et al., 2013). Fig. 8 represents the computational domain used for drift motion and rotating motion analyses. As shown in Fig. 8, the inlet, both sides, bottom, and top boundaries are located approximately 0.5L away from the bow of the hull, while the outlet boundary is positioned about 1L

away from the stern. Additionally, the boundary located at the stern aft is set as a pressure outlet, while the others are all set as velocity inlets for the analysis.

3.3. Grid convergency studies

To obtain reliable analysis results, grid convergence tests were conducted. Three types of grids—coarse, medium, and fine—were generated in the grid system. To verify grid convergence, the lateral force and moment acting on the bare hull of the DARPA Suboff in drift ($\beta = 12^\circ$) and turning ($r' = 0.3702$, $\beta = 16.5$) conditions were calculated for each grid system while maintaining a time step of 0.01 s. Table 6 provides detailed information on the grid systems used for the convergence analysis and the corresponding calculation results.

The grid convergence index (GCI) was calculated by using Eq. (19) to analyze the degree of grid convergence quantitatively (Celik et al., 2008).

$$GCI_{fine}^{21} = \frac{1.25 \left| \frac{Q_1 - Q_2}{Q_1} \right|}{r_{21}^p - 1} \quad \text{where } p = \frac{1}{\ln(r_{21})} \left| \ln \left| \frac{\epsilon_{32}}{\epsilon_{21}} \right| \right|, \quad \epsilon_{ij} = Q_i - Q_j \quad (27)$$

In Eq. (27), the subscripts denote the grid systems, where 1, 2, and 3 represent the coarse, medium, and fine grids, respectively. Additionally, r_{21} represents the ratio of grid size between the fine and medium grids, which has a value of 1.2. ϵ_{ij} represents the difference in the physical quantities (lateral force and moment) between grid systems i and j . Using the non-dimensional lateral force Y and non-dimensional moment N as target physical quantities, as shown in Table 6, the Grid Convergence Index (GCI) was calculated. For the drift condition ($\beta = 12^\circ$),

Table 6
Results of grid dependency study for both drift and rotation motion.

		Coarse	Medium	Fine	R	GCI
Base size (m)		0.24	0.2	0.1667	—	—
Cells		5,438,639	8,718,666	14,189,883	—	—
Drift motion ($\beta = 12^\circ$)	Y'	7.146E-03	7.141E-03	7.140E-03	0.212	0.0052%
	N'	2.013E-03	2.017E-03	2.020E-03	0.678	0.397%
	Y' error	—	0.073%	0.016%	—	—
	N' error	—	−0.223%	−0.151%	—	—
Rotation motion ($r' = 0.3702$)	Y'	1.500E-02	1.493E-02	1.493E-02	0.154	0.015%
	N'	1.057E-02	1.061E-02	1.061E-02	−0.031	0.005%
	Y' error	—	0.425%	0.065%	—	—
	N' error	—	−0.406%	0.012%	—	—

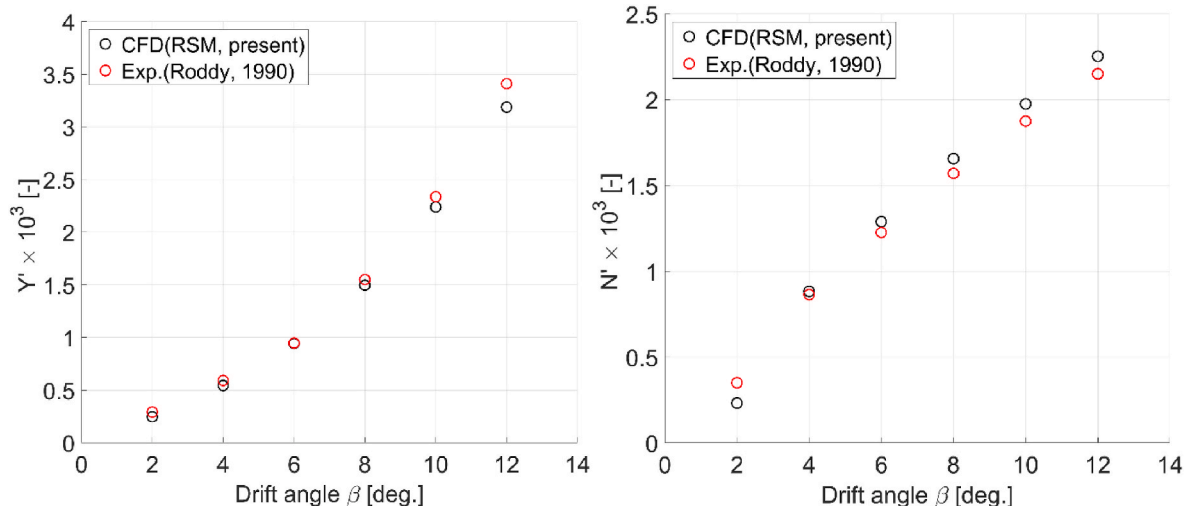


Fig. 9. Comparison results between CFD and experiment (lateral force and moment) at drift motion.

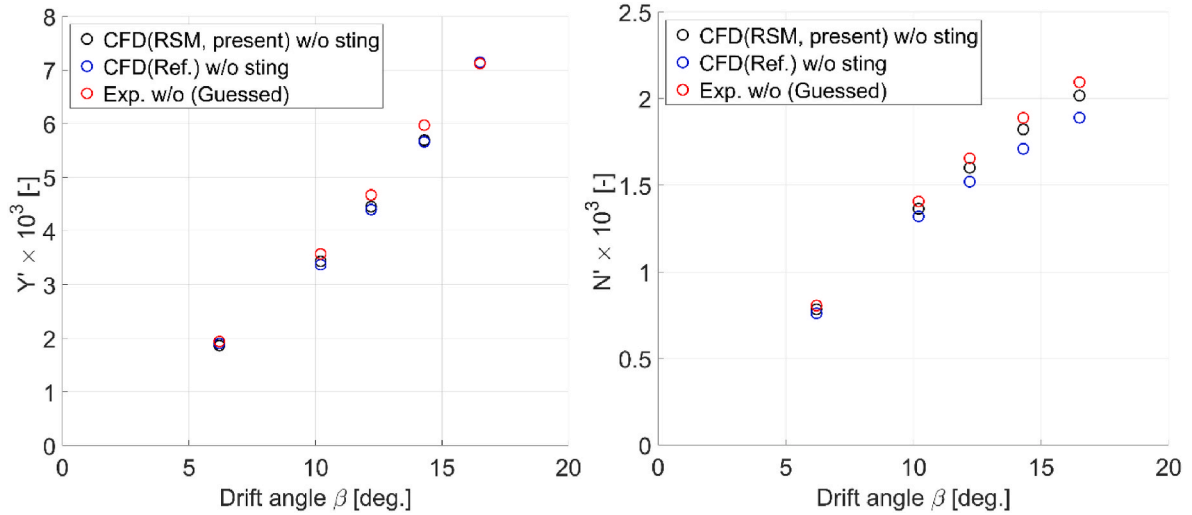


Fig. 10. Comparison results between CFD and experiment (lateral force and moment) at rotation motion with drift angle.

relative to the fine grid, the lateral force and moment have GCI values of 0.005% and 0.397%, respectively. For the turning condition ($r' = 0.37$; $\beta = 16.5^\circ$), the lateral force and moment have GCI values of 0.015% and 0.005%, respectively. GCI represents the uncertainty of a calculated physical quantity in a specific grid system. In both drift and turning motions, GCI shows uncertainty of less than 1% relative to the fine grid. Additionally, to examine the convergence based on grid size, Eq. (28) can be used (Xing and Stern, 2010).

$$R = \frac{\epsilon_{21}}{\epsilon_{32}} \begin{cases} 0 < R < 1 : \text{monotonic convergence} \\ -1 < R < 0 : \text{oscillatory convergence} \\ 1 \leq |R| : \text{divergence} \end{cases} \quad (28)$$

As summarized in Table 6, the non-dimensional lateral force Y and moment N for the drift condition have R values of 0.212 and 0.687, respectively. Similarly, for the turning condition, the non-dimensional lateral force Y and moment N have R values of 0.154 and -0.031 , respectively. From these values, it can be observed that the moment during turning motion exhibits oscillatory convergence, while in other cases, it shows monotonic convergence. Additionally, investigating the relative errors based on grid size reveals that both the lateral force and moment decrease in error as the grid size decreases for both drift and turning motions, demonstrating convergence even when using a medium grid resolution.

3.4. Results and comparison

Fig. 9 presents a comparison between the forces and moments acting on the hull as a function of drift angle, obtained from model tests and CFD analysis results. It can be observed from Fig. 9 that as the drift angle increases, the discrepancies between the values also increase. However, except for a drift angle of 2° , the results closely match within a maximum deviation of 10%.

Meanwhile, Fig. 10 presents the calculated forces and moments on the hull during steady turning as a function of the drift angle in rotating arm test. The red circles in Fig. 10 represent the predicted values from the study of Etebari et al. (2008), which reflect the trend of changes in the lateral force and moment acting on the Suboff hull with or without a sting, based on their CFD calculations. It can be observed from Fig. 10 that the predicted values from the model tests and CFD calculations closely match within a maximum deviation of 7% for both lateral force and moment.

Table 7
Principal dimensions of MARIN BB2.

Quantity	Symbol	Unit	Prototype	Model
Length from nose to tail	L_{OA}	[m]	70.200	3.826
Breadth	B	[m]	9.600	0.5232
Depth	D	[m]	10.600	0.578
Depth including sail	D_{Sail}	[m]	16.200	0.883
Displacement	∇	[m ³]	4365.2	0.719
Longitudinal center of gravity from nose	L_{CG}	[m]	32.31	1.761
Vertical center of gravity from keel	V_{CG}	[m]	0.0443	0.002
Propeller diameter	D_p	[m]	5.00	0.273
Scale ratio	—	—	1	18.348

4. Numerical captive calculations for validating the physical based manoeuvring mathematical model

This section describes a method for implementing large drift and/or angle of attack motions that cannot be considered with model testing facilities using computational fluid dynamics (CFD) and obtaining hydrodynamic force elements proposed in this study from the obtained forces. Although the accuracy of hydrodynamic forces obtained from CFD cannot be guaranteed, the aim of this study is to propose a mathematical model capable of simulating a wide range of 6DOF manoeuvring motions of submarines. Therefore, even if there are errors, the focus is on the method and procedure for obtaining hydrodynamic force elements to construct the mathematical model, acknowledging the errors. Additionally, within the range feasible for model testing, the validity of the CFD analysis is verified through a comparison of results between model testing and CFD analysis. Finally, by comparing the results of free-running model tests conducted by MARIN with BB2 as the

Table 8
Captive calculation conditions.

Analysis items (Captive calculation types)	Range
Horizontal static drift	$\beta = 0^\circ, 10^\circ, 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ, 70^\circ, 80^\circ, 90^\circ$
Vertical static angle of attack	$\alpha = 0^\circ, \pm 10^\circ, \pm 20^\circ, \pm 30^\circ, \pm 40^\circ, \pm 50^\circ, \pm 60^\circ, \pm 70^\circ, \pm 80^\circ, \pm 90^\circ$
Horizontal rotation with static drift angle (symmetry about vertical center plane)	$r' = 0.1, 0.2, 0.3, 0.4, 0.5$ at $\beta = 0^\circ, 10^\circ, 15^\circ$
Vertical rotation with static angle of attack (asymmetry about horizontal center plane)	$q' = 0.1, 0.2, 0.3, 0.4, 0.5$ at $\alpha = 0^\circ, 5^\circ, 10^\circ, 15^\circ$ $q' = -0.1, -0.2, -0.3, -0.4, -0.5$ at $\alpha = 0^\circ, -5^\circ, -10^\circ$

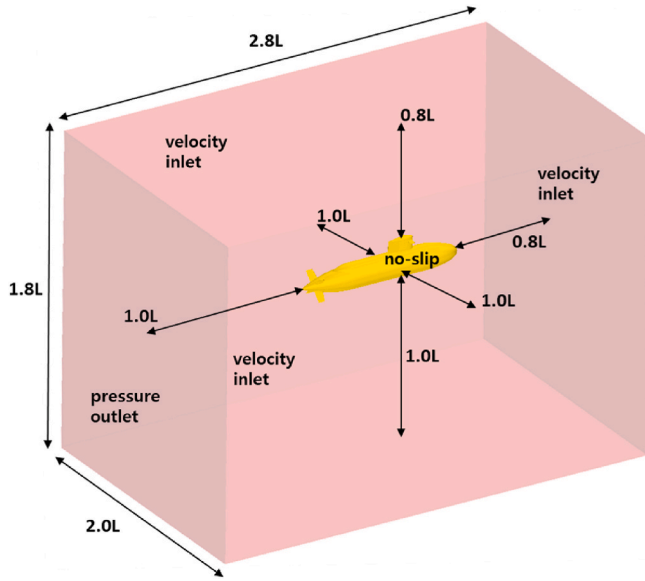


Fig. 11. Computational domain and boundary conditions.

target ship and the simulation results based on the proposed mathematical model in this study, the usefulness of the proposed mathematical model and the procedure and method proposed for constructing the mathematical model is presented in this section.

4.1. Setup for numerical simulations

The principal dimensions of the MARIN BB2 used in the CFD analysis (Overpelt et al., 2015) are presented in Table 7. Additionally, the fundamental numerical analysis settings applied in the analysis are consistent with those used in previous calculations for DARPA Suboff (refer to Table 6), and the calculation conditions, grid systems, and boundary conditions are shown in Table 8 and Fig. 11. Because the MARIN BB2 has several appendages such as sail, control surfaces and so on, the domain size for it was determined from Kwon et al. (2023) instead of following the domain size of DARPA Suboff case.

Taking into account the symmetry of the submarine hull, the analysis range for the horizontal drift analysis was set from 0° to 90° , considering the starboard and port side symmetry, and for the vertical pitch analysis, it was set from -90° to 90° , considering the top-bottom asymmetry. Additionally, for both horizontal and vertical turning motions, the analysis was performed only for the items indicated in Table 8. This was done to confirm the ability of the mathematical model proposed by Karasuno et al. (1991, 1992) to estimate the hydrodynamic forces acting on the hull during turning motions only based on the hydrodynamic forces obtained from drift motion.

4.2. Data analysis to obtain elementary hydrodynamic forces

Before obtaining the fluid force elements constituting the mathematical model (Eq. (13)–(18)) from the results of the CFD analysis for the items listed in Table 9, it is necessary to briefly explain the procedure of the analysis. Fig. 12 illustrates the procedure for obtaining the fluid force elements that constitute the mathematical model proposed in this study after obtaining the hydrodynamic forces (forces and moments) acting on the hull during drift motion from model tests or CFD analysis. While this study utilizes CFD analysis results, it is noted that if model test results are available, they can be used instead.

If model tests or CFD analyses have been conducted under given conditions, as shown in Fig. 12(b) and (c), first(①), from the longitudinal forces, lateral forces, and moments at a drift angle (or angle of attack) of 90° ($v' = 1.0$; $u' = 0.0$), determine C'_{LAS} (from Table 9, step 5),

Table 9

Analysis procedure to obtain the coefficient of all elementary components of hydrodynamic forces and moments (see Fig. 12).

Coefficients for elementary force and moment		Descriptions
1. Munk moment	m'_x, m'_y $N_{Munk} = (m'_x - m'_y)u'v'$	Obtain the added mass from the potential theory or the experiment. N_{Munk} can be obtained.
2. Crossflow drag ① in Fig. 12	$C'_{D90}, C'_{D90} \cdot m'_0, C'_{D90} \cdot m'_1$ $Y_c = \frac{1}{2} \rho L_{pp}^2 U^2 C'_{D90} m'_0 v'^3$ $N_c = \frac{1}{2} \rho L_{pp}^2 U^2 C'_{D90} m'_1 v'^3$	Determine the coefficients at $\beta = 90^\circ$ from the calculations (or experiments) for the lateral force and the moment (Y and N). Y_c and N_c can be obtained.
3. Lift ② in Fig. 12	$C'_L, C'_L \cdot l'_{v0}$ $X_{Lv} = \frac{1}{2} \rho L_{pp}^2 U^2 C'_L u' v'^2$ $Y_{Lv} = \frac{1}{2} \rho L_{pp}^2 U^2 C'_L u' v'$ $N_{Lv} = \frac{1}{2} \rho L_{pp}^3 U^2 C'_L l'_{v0} u' v'^2$	After removing the known terms ($C'_{D90}, C'_{D90} \cdot m'_0, C'_{D90} \cdot m'_1$) from the calculations (or experiments) for Y and N , obtain the coefficients by using least square method at $\beta = 0^\circ \sim 30^\circ$. X_{Lv} , Y_{Lv} and N_{Lv} can be obtained.
4. Drag ③ in Fig. 12	$C'_D, C'_{Di} \cdot l'_{i0}, p$ $X_{Di} = \frac{1}{2} \rho L_{pp}^2 U^2 C'_{Di} u'^3 v'^2$ $Y_{Di} = \frac{1}{2} \rho L_{pp}^2 U^2 C'_{Di} u'^2 v'^2$ $N_{Di} = \frac{1}{2} \rho L_{pp}^3 U^2 C'_{Di} l'_{i0} u'^2 v'^3$ $p \cdot \cos^2 \beta$	After removing the known terms ($C'_{D90}, C'_{D90} \cdot m'_0, C'_{D90} \cdot m'_1, C'_L, C'_L \cdot l'_{v0}$) from the calculations (or experiments), obtain the coefficients by using least square method at $\beta = 0^\circ \sim 90^\circ$. X_{Di} , Y_{Di} and N_{Di} can be obtained.
5. Crossflow lift & Friction ① & ④ in Fig. 12	C'_F, C'_{LAS} $X_F = \frac{1}{2} \rho L_{pp}^2 U^2 C'_F u' u' $ $X_{LAS} = \frac{1}{2} \rho L_{pp}^2 U^2 C'_{LAS} v'^2$	Determine C'_F from the longitudinal force (X) at $\beta = 0^\circ$ and remove all related terms obtained from the previous steps and obtain C'_{LAS} from the longitudinal force (X) at $\beta = 90^\circ$. X_F and X_{LAS} can be obtained.
6. Correction force for stall effect	$\Delta C'_L, \Delta C'_{Di}$ $(L'_{ST} + D'_{ST})_{XF} = -\Delta$ $C'_L u' v'^2 + \Delta C'_{Di} u'^3 v'^2$	After removing all related known terms from the calculations for X , obtain the coefficients by using the least square method at $\beta = 0^\circ \sim 90^\circ$. X_{ST} can be obtained.
7. Total force & moment Final result in Fig. 12	$X = X_{Lv} + X_{Di} + X_{LAS} + X_{ST} + X_F$ $Y = Y_{Lv} + Y_{Di} + Y_C$ $N = N_{Lv} + N_{Di} + N_C + N_{Munk}$	

$C'_{D90} \cdot m'_0$, and $C'_{D90} \cdot m'_1$. Next(②), use the least squares method to determine C'_L for the values in the range of $0^\circ \sim 30^\circ$ of drift angle (or angle of attack), after removing the known component Y_c (refer to step 2 in Table 9) from the CFD results, as shown in Fig. 12(b). Then(③), apply the least squares method to determine C'_{Di} and p for the values in the range of $0^\circ \sim 90^\circ$ drift angle (or yaw angle) after removing the known component Y_{Lv} (refer to step 3 in Table 9) from the CFD result, as in step ② (for the vertical angle of attack motion case, use the range $-90^\circ \sim 90^\circ$). Additionally(④), determine C'_F for the longitudinal force at a drift angle (or angle of attack) of 0° . Following that(⑤), use the least squares method to determine $C'_L - \Delta C'_L$ and $C'_{Di} - \Delta C'_{Di}$ after removing the known components X_F and X_{LAS} (refer to step 5 in Table 9) from the longitudinal force for the values in the range of $0^\circ \sim 90^\circ$ drift angle (or angle of attack). Then, remove the obtained C'_{Di} and C'_{Di} to determine $\Delta C'_L$ and $\Delta C'_{Di}$ and obtain $(L'_{ST} + D'_{ST})$. Finally(⑥), determine $C'_L \cdot l'_{v0}$ for the range of $0^\circ \sim 30^\circ$ drift angle (or angle of attack) using the least squares method after removing all known terms (N_{Munk} and N_c in Table 9) from the moment. Similarly, remove the known terms again and use the least squares method to determine $C'_{Di} \cdot l'_{i0}$ for the range of $0^\circ \sim 90^\circ$ drift angle (or angle of attack) (for the vertical angle of attack motion case, use the range $-90^\circ \sim 90^\circ$).

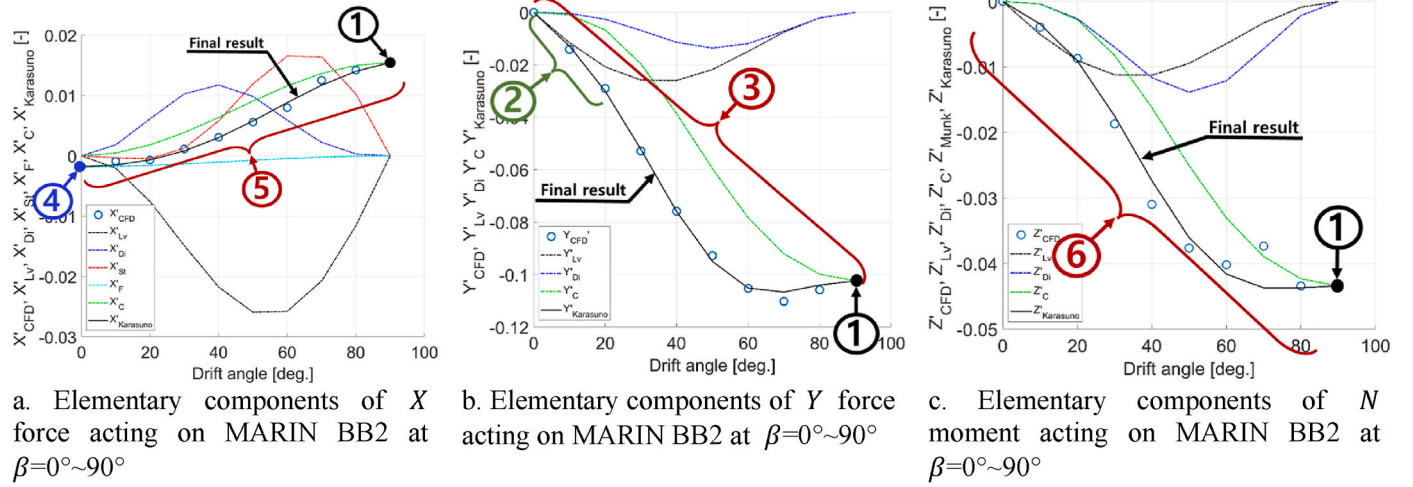


Fig. 12. Analysis procedure to obtain elementary components of hydrodynamic forces.

Table 10

Results of grid dependency study for both drift and rotation motion.

		Coarse	Medium	Fine	R	GCI
Base size (m)		0.2828	0.2	0.1414	–	–
Cells		3,054,895	6,157,406	13,286,652	–	–
Drift motion	Y'	1.697E-03	1.746E-03	1.767E-03	0.425	1.097%
$(\beta = 12^\circ)$	N'	3.454E-03	3.248E-03	3.220E-03	0.136	0.172%
	Y' error	–	2.903%	–1.186%	–	–
	N' error	–	–5.957%	0.871%	–	–

4.3. Grid convergency studies and validation

For the CFD calculations for MARIN BB2, grid convergence studies were conducted following the same procedure as in Section 3.3, similar to the CFD analysis conducted for DARPA Suboff. In addition, according to the results of the temporal convergence studies by Kwon et al. (2023), because the time step under 0.02 can give reasonable results, a time step of 0.01 s is used in the study without the convergence studies for the temporal discretization.

Table 10 provides detailed information on the grid configurations and results used for the convergence analysis.

As summarized in Table 10, the dimensionless lateral force Y and moment N at the drift angle exhibit R values of 0.425 and 0.136, respectively. Both lateral force and moment show monotonic convergence from these values. Furthermore, investigation of relative errors based on grid size reveals that as the grid size decreases, errors decrease for both lateral force and moment, indicating convergence. Additionally, even using a moderate grid size results in converged outcomes.

Moreover, for validation of the computations, comparisons were made with previous CFD studies and model test results. These previous analyses and model tests were conducted over the range of drift angles and yaw angles from -19° to 19° . The model test conducted by KRISO was performed with a propeller installed, while the other CFD results (Bettle, 2014; Cho et al., 2020; Kwon et al., 2023; Mai et al., 2022; Toxopeus, 2017) were obtained under bare hull conditions. Fig. 13 compares various CFD and present results of the non-dimensional Y force and N moment depending on the horizontal static drift angle, showing that both sets of results are consistent. In addition, Fig. 14 presents a comparison of the experiment and CFD results for both horizontal static drift and vertical angle of attack conditions. These results confirmed that the present CFD results were in good agreement with the experiment results. Therefore, it is thought that the present CFD results for both the horizontal static drift and the vertical angle of attack can be

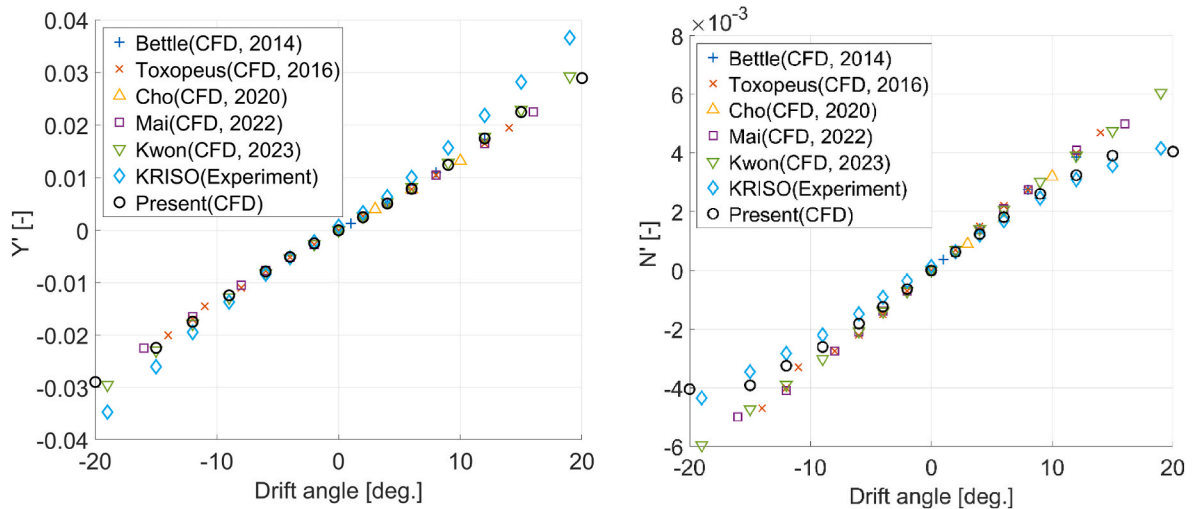


Fig. 13. Comparison of various CFD results for horizontal static drift conditions.

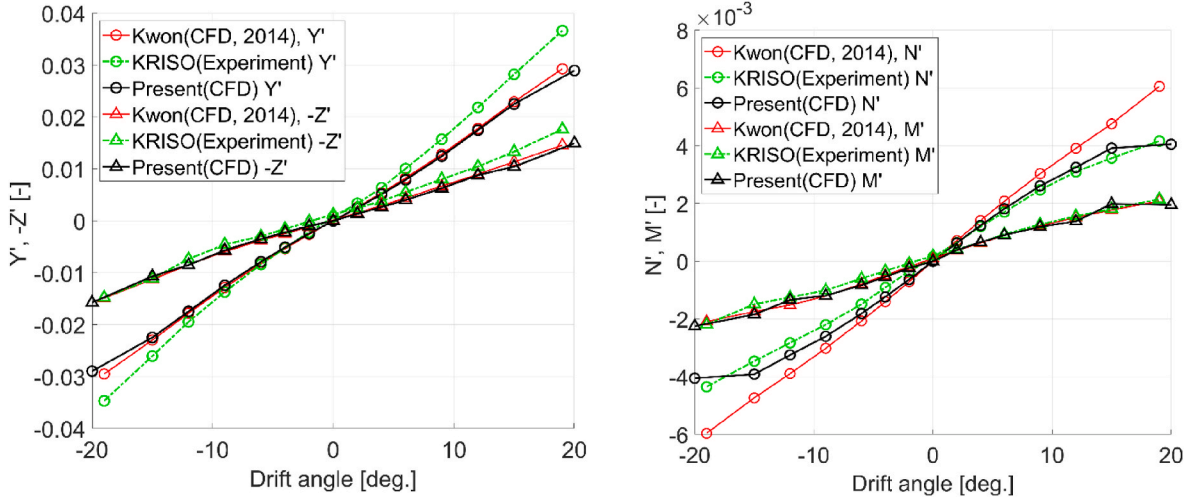


Fig. 14. Comparison of experiment and CFD results for both horizontal static drift and vertical angle of attack conditions.

used for the proposed mathematical model in this study.

4.4. Analysis results and discussion

4.4.1. Horizontal static drift

Fig. 15 illustrates the results of applying the mathematical model proposed in this study to analyze the forces and moments acting on the submarine under horizontal static drift conditions ($\beta = 0^\circ \sim 90^\circ$). The results in (a), (b), and (f) depict the horizontal forces and moments acting on the submarine during horizontal static drift motion, while those in (c), (d), and (e) represent the vertical forces and moments generated during horizontal plane motion. These vertical forces and moments are components caused by the asymmetrical shape of the submarine hull in the vertical direction. Fig. 15 shows that the mathematical model (Karasuno's model) proposed in this study effectively identifies the forces and moments acting on the submarine during the horizontal drift motion. Other colored lines except the black solid line present elemental component forces, as shown in Fig. 15. In addition, the black solid lines in Fig. 15 (a) ~ (f) are reconstructed non-dimensional forces and moments, respectively.

Fig. 16 presents the flow structure around the submarine at drift angles of 30° and 80° . In Fig. 16 (a), cross flow developed from the submarine's bow keel extends around the submarine, forming vortices that flow towards the bow. Vortices originating from the submarine's sail are also observed, as well as vortices generated from the stern x-type rudder. On the other hand, Fig. 16 (b) shows the complex flow structure around the submarine's hull; a collapsed flow structure without any specific flow pattern is observed throughout the hull compared to Fig. 16 (a), with a cross flow around the hull. Due to these flow differences, the trend of lateral force shown in Fig. 15 (b) exhibits an increase in lateral force until around a 70° drift angle, followed by a decrease near 80° .

Fig. 17 compares results based on Karasuno's model with the results based on Taylor expansion. In Fig. 17(a) and (b), the horizontal axis represents the horizontal drift angle (β) and the vertical axis represents the non-dimensional lateral force and yaw moment; the horizontal axis in Fig. 17 (c) and (d) represents the non-dimensional lateral velocity (v) obtained from its definition ($\beta = -\sin^{-1}(\frac{v}{V})$). Fig. 17 (a) and (b) were added for convenience of explanation. In Fig. 17, the black solid line is the reconstructed non-dimensional force and moment from the coefficients (C_L^H, C_D^H, C_{i0}^H and C_{D90}^H) analyzed with Karasuno's model (Eqs. (22) and (26)) and both the red and the green solid lines are reconstructed non-dimensional force and moment from the coefficients ($Y_v, Y_{v|v|}, N_v$ and $N_{v|v|}$) analyzed by using the least square with Taylor 3rd order approximation model ($Y' = Y_v v' + Y_{v|v|} v'|v'|$; $N' = N_v v' + N_{v|v|} v'|v'|$).

Hereafter the results denoted by a black solid line are referred to as case 1, the results with a red solid line are called case 2, and the results with a green solid line are case 3. The difference between case 2 and case 3 derives from the data range when the least square method is applied to obtain the hydrodynamic coefficients from the CFD results. The data range for case 2 is from $\beta = -90^\circ \sim 90^\circ$ and for case 3 it is $\beta = -30^\circ \sim 30^\circ$. As shown in Fig. 17(a) and (b), case 1 (Karasuno's model) agrees with the CFD results. In contrast to case 1, case 2 (reconstructed results using coefficients obtained from $\beta = -30^\circ \sim 30^\circ$) agrees with the CFD results in a drift angle $\beta = -40^\circ \sim 40^\circ$ and the difference becomes larger at $|\beta| \geq 50^\circ$. However, case 3 (reconstructed results using coefficients obtained from $\beta = -90^\circ \sim 90^\circ$) does not agree with the CFD results in the drift angle $\beta = -90^\circ \sim 90^\circ$ and the trend is similar to the CFD results.

Therefore, Fig. 17 indicates that the Taylor based mathematical model has limitations for large drift angles motion and must have higher-order terms, as previously mentioned, to overcome these limitations. In addition, it is thought that Karasuno's model can be applied to large drift angle motion without modifying its form.

4.4.2. Vertical static angle of attack

Fig. 18 illustrates the results of the analysis applying the mathematical model proposed in this study for vertical angle of attack conditions ($\alpha = -90^\circ \sim 90^\circ$). From Fig. 18, it can be observed that the forces and moments during descent motion ($-90^\circ \leq \alpha \leq 0^\circ$) and ascent motion ($0^\circ \leq \alpha \leq 90^\circ$) present asymmetry. Furthermore, it is evident that while Karasuno's model accurately represents the forces and moments during ascent motion ($0^\circ \leq \alpha \leq 90^\circ$) in Fig. 18(a)–(c) and (e), there are differences in the large angle of attack range ($-90^\circ \leq \alpha \leq -40^\circ$) for descent motion ($-90^\circ \leq \alpha \leq 0^\circ$) in Fig. 18(a)–(c) and (e). This shows limitations in accurately capturing the asymmetry of forces and moments arising from the vertical asymmetry of the submarine's hull. Results (a), (c), (d), and (e) in Fig. 18 represent the vertical forces and pitch moments during vertical angle of attack motion, while (b) and (f) depict the lateral forces and yaw moments during vertical plane motion. These lateral forces and yaw moments are attributed more to the gradual development of the wake originating from the submarine's hull rather than the coupling effect due to hull asymmetry, as the submarine hull is symmetric about the port/starboard axis. Comparing the magnitudes of forces and moments shown in Fig. 15 (b) and (f) with those in Fig. 18 (b) and (f), it can be observed that the coupling forces in Fig. 18 are relatively small. Fig. 18 demonstrates that the mathematical model proposed in this study effectively represents the forces and moments acting on the submarine during both horizontal and vertical plane large angle motion.

Fig. 19 compares results based on the Karasuno's model with the results based on Taylor expansion. In Fig. 19 (a) and (b), the horizontal

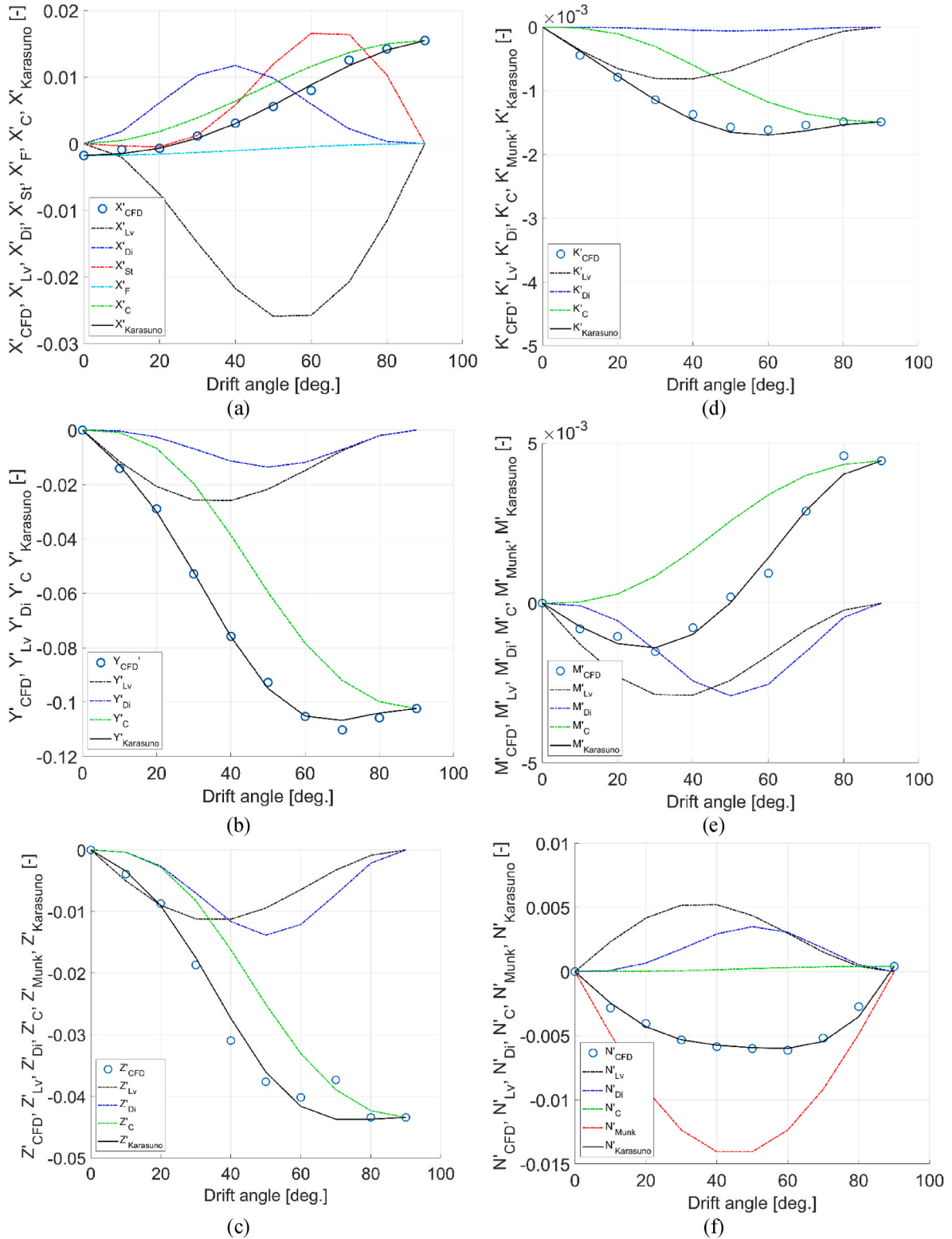


Fig. 15. Analyzed results of hydrodynamic forces (X' , Y' , Z') and moments (K' , M' , N') at horizontal static drift conditions by proposed mathematical model. (a) Longitudinal force, (b) Lateral force, (c) Vertical force, (d) Roll moment, (e) Pitch moment, (f) Yaw moment.

axis represents the vertical angle of attack(α) and the vertical axis represents the non-dimensional vertical force and moment; the horizontal axis in Fig. 19 (c) and (d) represents the non-dimensional vertical velocity (w'). Fig. 19 (a) and (b) were also added for convenience of explanation. In Fig. 19, case 1 is the reconstructed non-dimensional force and moment from the coefficients (C_L' , C_{Di}' , C_{i0}' and C_{D90}')

analyzed with Karasuno's model (Eqns. (23) and (25)) and both case 2 and case 3 are the reconstructed non-dimensional force and moment from the coefficients (Z_w , $Z_{w|w|}$, M_w and $M_{w|w|}$) analyzed by using least square with Taylor 3rd order approximation model ($Z' = Z_w w' + Z_{w|w|} w'|w'|$; $M' = M_w w' + M_{w|w|} w'|w'|$). The results in Fig. 19 are obtained with same procedure mentioned in section 4.4.1. As shown in Fig. 19 (a) and

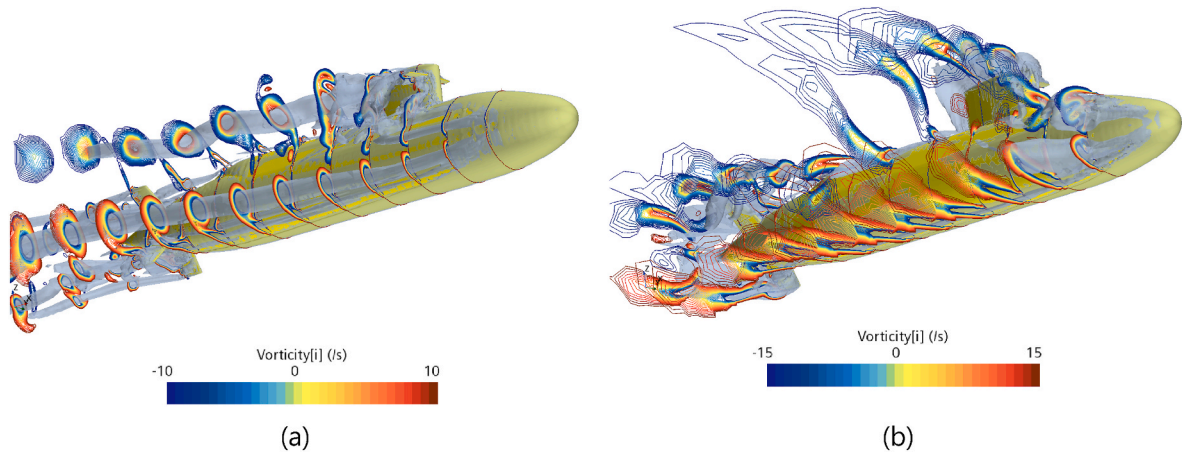


Fig. 16. Comparison of flow structure at horizontal static drift conditions. (a) drift angle = 30°, (b) drift angle = 80°.

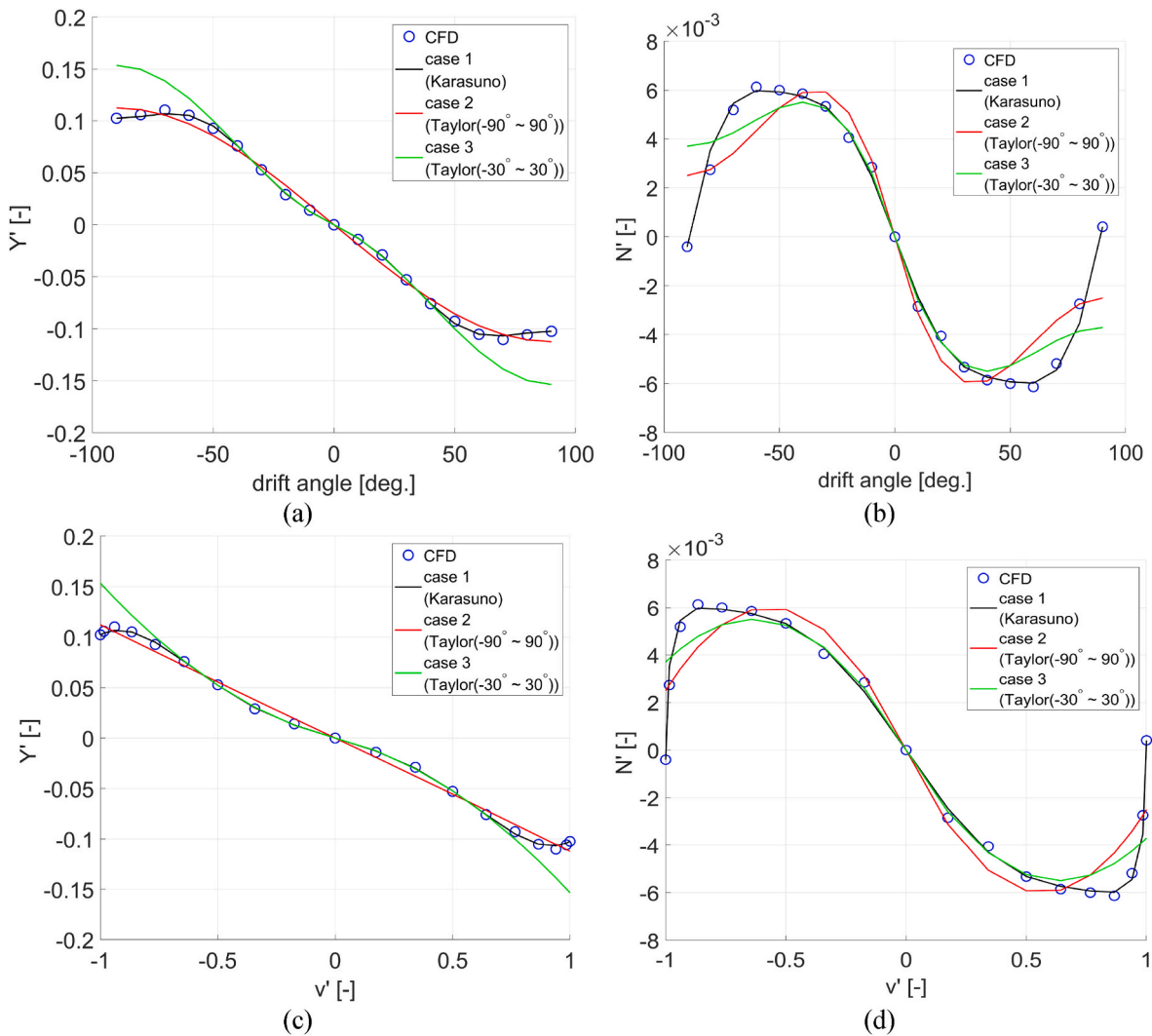


Fig. 17. Comparison of Karasuno model and Taylor based model for lateral force and yaw moment.

(b), case 1 (Karasuno's model) agrees with the CFD results except in the angle of attack $\alpha = -30^\circ \sim -60^\circ$. In contrast to case 1, case 2 agrees with the CFD results in the angle of attack $\alpha = -50^\circ \sim 50^\circ$ and the difference becomes larger at $|\alpha| \geq 55^\circ$. For case 3, the same tendency as the results in section 4.4.1 is observed.

As mentioned in section 4.4.1, the same tendency as observed for the horizontal static drift motion is seen for the vertical static drift motion from Fig. 19.

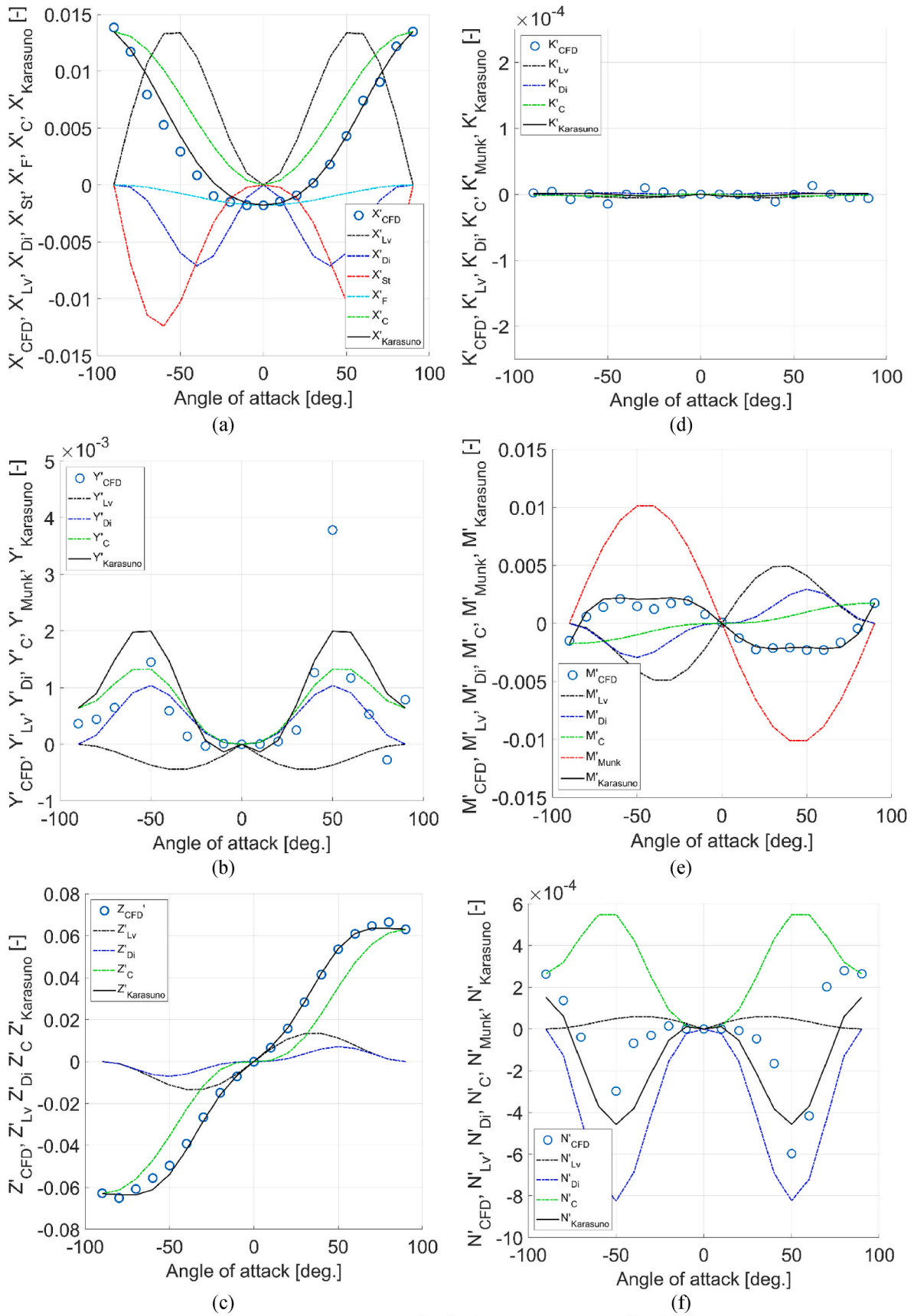


Fig. 18. Analyzed results of hydrodynamic forces (X' , Y' , Z') and moments (K' , M' , N') at vertical angle of attack conditions by proposed mathematical model. (a) Longitudinal force, (b) Lateral force, (c) Vertical force, (d) Roll moment, (e) Pitch moment, (f) Yaw moment.

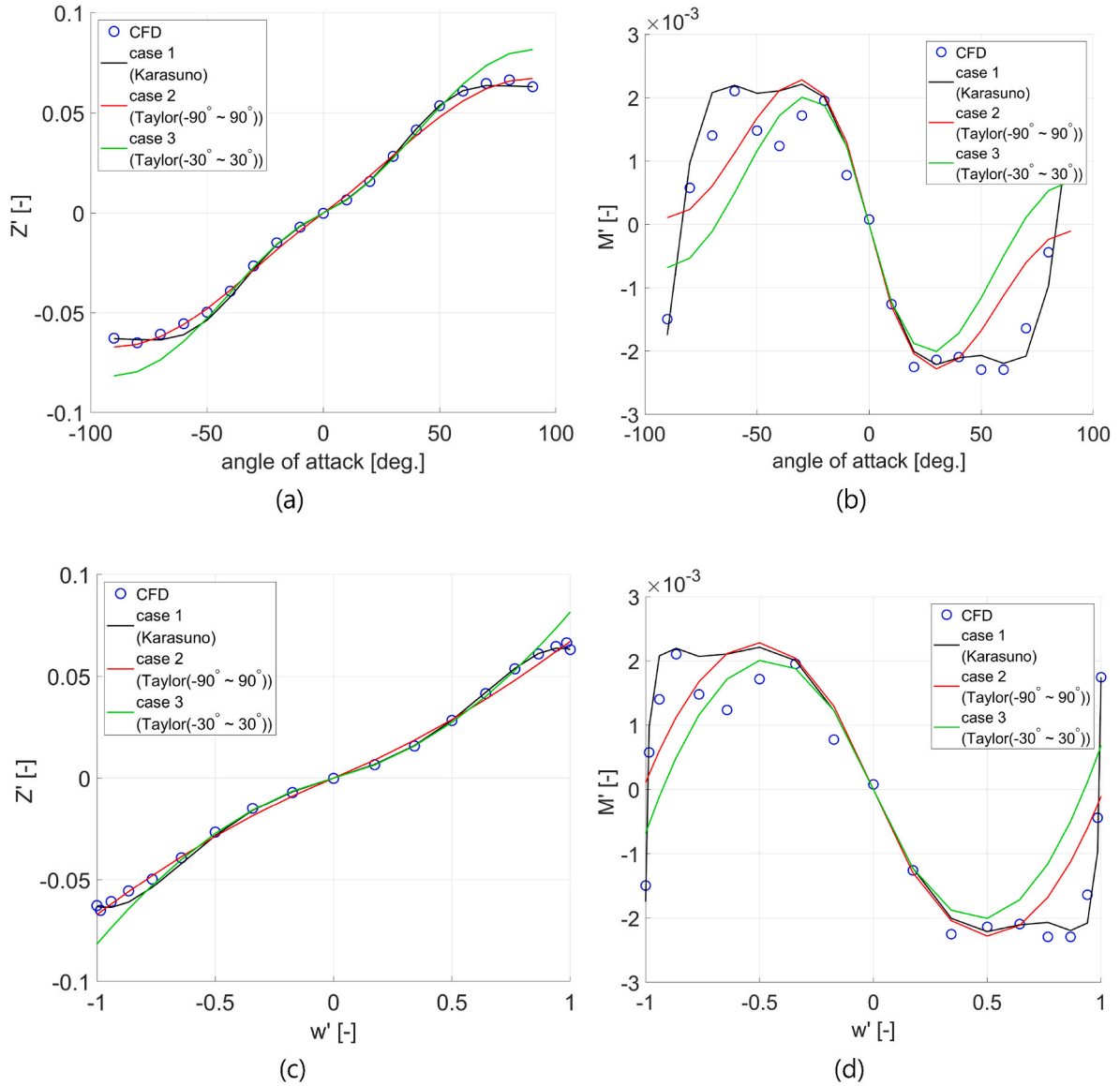


Fig. 19. Comparison of Karasuno model and Taylor based model for vertical force and pitch moment.

Table 11

Manoeuvring hydrodynamic coefficients obtained from Taylor based mathematical model.

Hydrodynamic derivatives for horizontal motion	Values	Hydrodynamic derivatives for horizontal motion	Values
Y_v	-0.0557	Z_w	-0.02761
$Y_{v v }$	-0.0978	$Z_{w w }$	-0.05398
Y_r	-0.0086	Z_q	0.01125
$Y_{r r }$	-0.0019	$Z_{q q }$	-0.00151
N_v	-0.0173	M_w	0.00871
$N_{v v }$	0.0136	$M_{w w }$	-0.00939
N_r	0.0058	M_q	0.00569
$N_{r r }$	0.0009	$M_{q q }$	-0.00022
$Y_{v r }$	0.0481	$Z_{w q }$	0.289
$Y_{v r }$	0.1828	$Z_{w q }$	0.1216
$N_{v r }$	0.0248	$M_{w q }$	0.01274
$N_{v r }$	0.0136	$M_{w q }$	0.01207

4.4.3. Horizontal rotation motion with static drift and vertical rotation motion with angle of attack

Consideration of forces and moments acting on the hull during the turning motions (yaw and pitch) on both the horizontal and vertical planes is necessary for submarines. To address this, in this study, CFD calculations for the turning motions with varying drift angle for the horizontal plane and angle of attack for the vertical plane were conducted separately. Table 11 presents the hydrodynamic coefficients obtained by applying the Taylor 3rd order approximation based mathematical model to CFD results obtained from the calculations specified in Table 9. It should be noted that these hydrodynamic coefficients are used to reconstruct the results shown in Figs. 20 and 21.

In Fig. 20 and 21, all solid lines present the reconstructed results from Karasuno's model and all dotted lines present the reconstructed results from the Taylor based model. Figs. 20 and 21 compare the CFD results and the reconstructed results from the Taylor-based model and Karasuno's model for the conditions listed in Table 9. In Fig. 20 (b) and 21 (b), the results when the drift angle (β) and angle of attack (α) are 0 (black solid line) represent pure rotational motion without the influence of lateral velocity. From Figs. 20 and 21, it can be observed that the Taylor based model shows increasing discrepancies with the CFD results

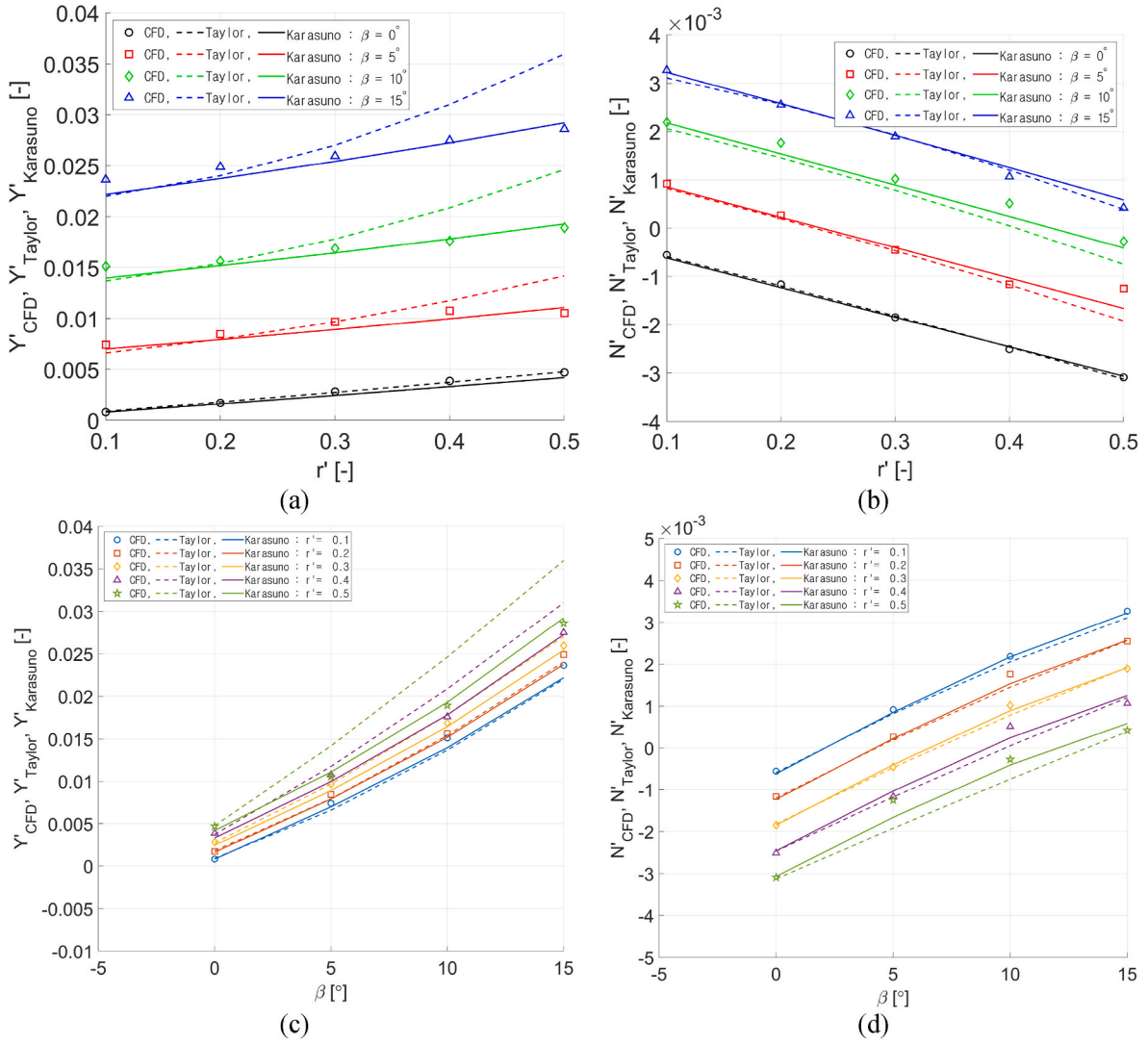


Fig. 20. Non-dimensional lateral force and yaw moment at horizontal rotation motion with static drift angle. (a) non-dimensional lateral force at fixed drift angle ($\beta = 0^\circ, 5^\circ, 10^\circ$ and 15°) with various non-dimensional angular velocities (r'), (b) non-dimensional yaw moment at fixed drift angle ($\beta = 0^\circ, 5^\circ, 10^\circ$ and 15°) with various non-dimensional angular velocities (r'), (c) non-dimensional lateral force at fixed non-dimensional angular velocity ($r' = 0.1, 0.2, 0.3, 0.4, 0.5$) with various static drift angles (β), (d) non-dimensional yaw moment at fixed non-dimensional angular velocity ($r' = 0.1, 0.2, 0.3, 0.4, 0.5$) with various static drift angles (β).

as the drift angle increases and the angular velocity of rotation increases. This is attributed to the effect of increased effective drift angle (or angle of attack) due to the increased angular velocity and the inability to adequately reflect the CFD calculation results used to obtain the hydrodynamic coefficients under various conditions. In fact, the Taylor based model has limitations dependent on the number of data points used for obtaining the hydrodynamic coefficients, as it employs curve fitting using the least squares method. On the other hand, the Karasuno model utilizes element coefficients obtained from the calculations of drift motion on the horizontal plane and the calculations of angle of attack motion on the vertical plane to compute the hydrodynamic forces and moments related to the turning motion without the need for curve fitting using the least squares method. However, determining variables such as c'_{v0} , x'_i , and x'_l is necessary for this purpose. According to Karasuno et al. (1991, 1992), while these variables are automatically determined for ships, they must be determined for submarines. This means that these parameters should be determined by a trial and error method. This is the limitation of Karasuno's model when it is applied to other vessels, especially submarines, as in this study.

As mentioned above, a separate process was conducted to determine these values to reconstruct the CFD results at a drift angle (β) and angle

of attack (α) of 0. From Figs. 20 and 21, it can be observed that the Karasuno model generally agrees well with the CFD calculation results except for under some conditions. This suggests that the proposed Karasuno model can be sufficiently used for the calculation of 6-DOF motions of submarines by accurately estimating the forces and moments associated with turning motion combined with drift angle (or angle of attack).

Fig. 20 (a) and (b) indicate that the non-dimensional lateral force and yaw moment become larger when the submarine is turning with varying drift angles. In this situation, the reconstructed force and moment from Karasuno's model agree with the CFD results while the results from Taylor based model do not agree with the CFD results at a large drift angle. Fig. 20 (c) and (d) present the same results with a different point of view. The same tendency is observed.

Fig. 21 (a) and (b) show that the non-dimensional vertical force and pitch moment become larger when the submarine is turning with varying angles of attack. In this situation, the reconstructed force and moment from Karasuno's model agree with the CFD results while the results from the Taylor based model do not agree with the CFD results at a large angle of attack. For Fig. 21(c) and (d), the same tendency is observed, as discussed previously.

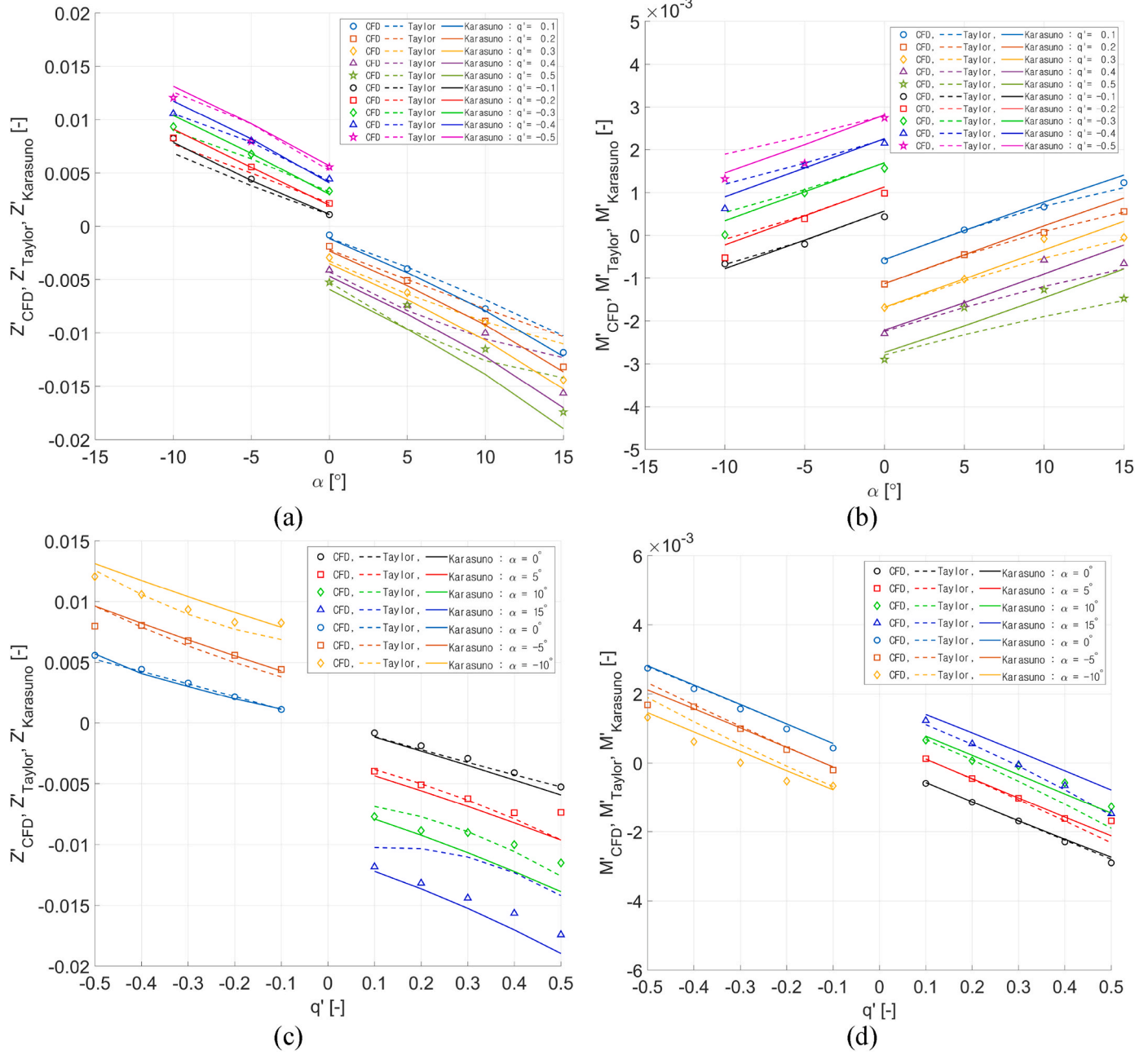


Fig. 21. Non-dimensional vertical force and pitch moment at vertical rotation motion with angle of attack angle. (a) non-dimensional vertical force at fixed angle of attack ($\alpha = 0^\circ, 5^\circ, 10^\circ$ and 15°) with various non-dimensional angular velocities (q'), (b) non-dimensional yaw moment at fixed drift angle ($\alpha = 0^\circ, 5^\circ, 10^\circ$ and 15°) with various non-dimensional angular velocities (q'), (c) non-dimensional vertical force at fixed non-dimensional angular velocity ($q' = 0.1, 0.2, 0.3, 0.4, 0.5$) with various angles of attack (α), (d) non-dimensional yaw moment at fixed non-dimensional angular velocity ($q' = 0.1, 0.2, 0.3, 0.4, 0.5$) with various static drift angles (α).

For the purpose of using Karasuno's model to estimate the forces and moments acting on the submarine's hull, the applicability of Karasuno's model was verified in sections 4.4.1 to 4.4.3. In section 4.4.1, it was found that in estimating the force and moment acting on the hull in the horizontal motion, Karasuno's model could give results with higher accuracy over a wider range of motion than the Taylor based model. Additionally, in Section 4.4.2, it was found that the same results were obtained even under a vertical angle of attack motion. Finally, in Section 4.4.3, it can be seen that the estimation results of the hydrodynamic force and moment acting on the hull in combined turning motion with the drift angle and/or angle of attack in the horizontal and vertical planes also produced estimation results with higher accuracy than the Taylor based model. From this, it was confirmed that Karasuno's model

is sufficiently applicable to estimate the maneuvering performance of the submarine considered in this study.

5. Numerical simulations

To validate the applicability of the mathematical model proposed in this study, numerical simulations were conducted and compared with free-running model tests performed by Overpelt et al. (2015). Hydrodynamic forces (F_C) for the control surfaces and thrust were obtained using the results from MARIN and Kim et al. (2021). The submarine's speed and major specifications were simulated under the same conditions as in Overpelt et al. (2015). The coefficients related to fluid forces used in the subsequent simulations are listed in Table 12.

Table 12
Hydrodynamic coefficients obtained from Karasuno mathematical model.

Hydrodynamic derivatives for horizontal motion	Values	Hydrodynamic derivatives for horizontal motion	Values
C_L^h	-0.1191	C_L^v	0.0331
$C_{D_i}^h$	-0.109	$C_{D_i}^v$	0.0356
ΔC_L^h	-0.142	ΔC_L^v	0.0633
$\Delta C_{D_i}^h$	-0.1444	$\Delta C_{D_i}^v$	0.0114
$C_L^h \tau_{v0}$	0.0119	$C_L^v \tau_{v0}$	0.0111
$C_{D_i}^h \tau_{v0}$	0.0098	$C_{D_i}^v \tau_{v0}$	0.02111
C_{LAS}^h	-0.006	C_{LAS}^v	-0.0129
C_F^h	0.0019	C_F^v	0.00019
$C_{D90}^h m'_0$	-0.2021	$C_{D90}^v m'_0$	0.1124
$C_{D90}^h m'_1$	-0.001	$C_{D90}^v m'_1$	0.0018

Table 13
Comparison results for turning circle manoeuvring motion.

	Overpelt et al. (2015)	Present	Difference $\left(\frac{\text{Overpelt} - \text{Present}}{\text{Overpelt}} \times 100 \right) [\%]$
Advance	2.75L	2.760L	-0.4%
Transfer	1.60L	1.763L	-10.2%
Tactical Diameter	3.40L	3.815L	-12.2%

5.1. Horizontal turning circle manoeuvring simulation

The results of the horizontal turning circle manoeuvring simulation with effective rudder angle $\delta_r = -20^\circ$ were compared with the model tests conducted by Overpelt et al. (2015), as shown in Fig. 21 and Table 13. Fig. 22 (a) presents a comparison of the submarine's turning trajectory, showing good agreement between the model test and numerical simulation results until the heading angle reaches 90° . However, discrepancies emerge afterward. Additionally, as presented in Fig. 22 (b), a decrease in ship speed starts deviating from the model test at around 12 s into the simulation (at approximately drift angle $\beta = 10^\circ$). This discrepancy is attributed to differences between the CFD calculations and model tests regarding the longitudinal force acting on the hull, as well as differences in the restoration of hull forces, specifically those attributable to the rudder based on the coefficients obtained from model tests conducted by Kim et al. (2021). Fig. 22 (c) illustrates a comparison of the yaw velocity, showing reasonable agreement with the model test results, despite some variation. Furthermore, the comparison results for advance and tactical diameter presented in Table 13 demonstrate that the values are within a maximum deviation of 13% compared to the model tests.

5.2. Horizontal zig-zag manoeuvring simulation

The results of the horizontal 10/10 zig-zag simulation were compared with those of the model test conducted by Overpelt et al. (2015) and are presented in Fig. 23 and Table 14. Fig. 23 (a) compares the time series of the submarine's heading angle and effective horizontal rudder angle, while Fig. 23 (b) and (c) show the time series of changes in

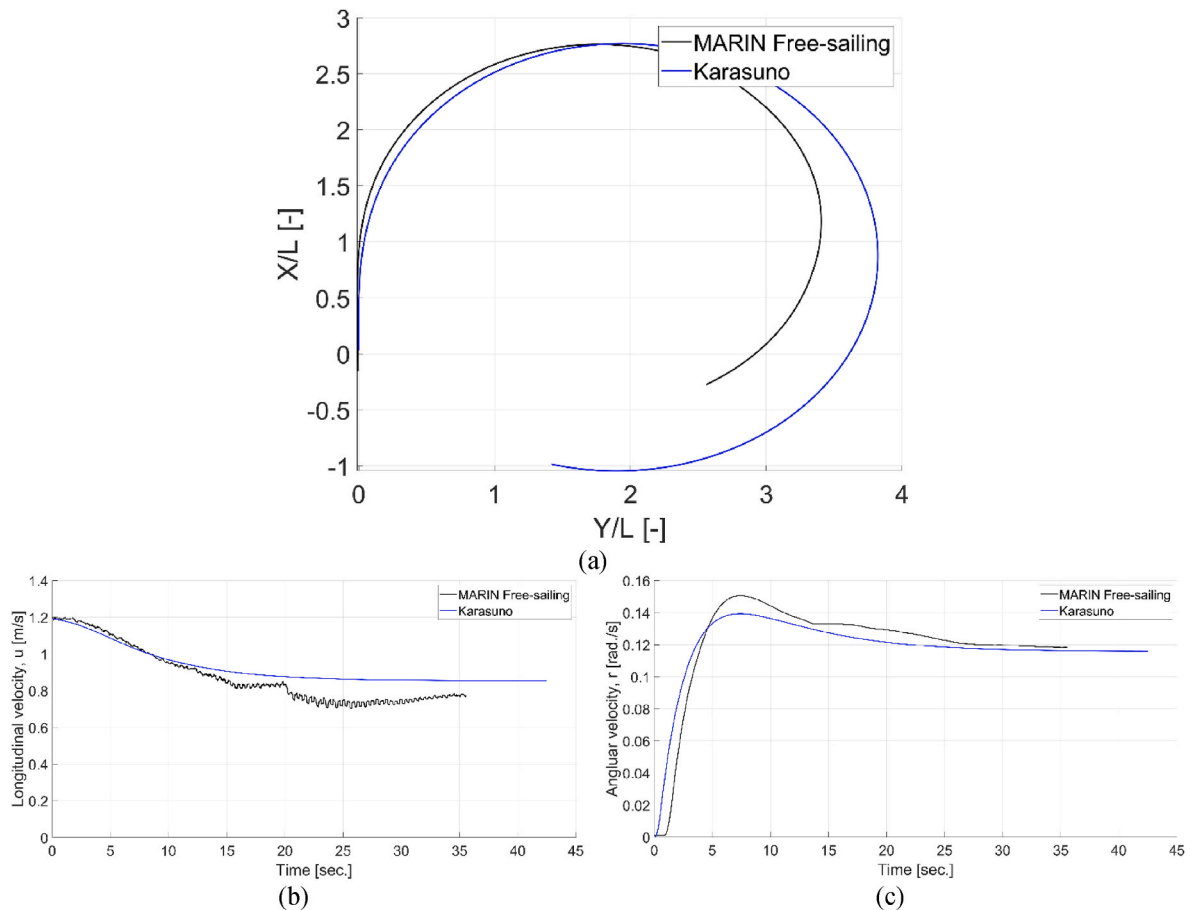


Fig. 22. Comparison results with free-sailing model test for turning circle manoeuvring: (a) Turning trajectory, (b) Time histories for longitudinal velocity(u), (c) Time histories for yaw velocity(r).

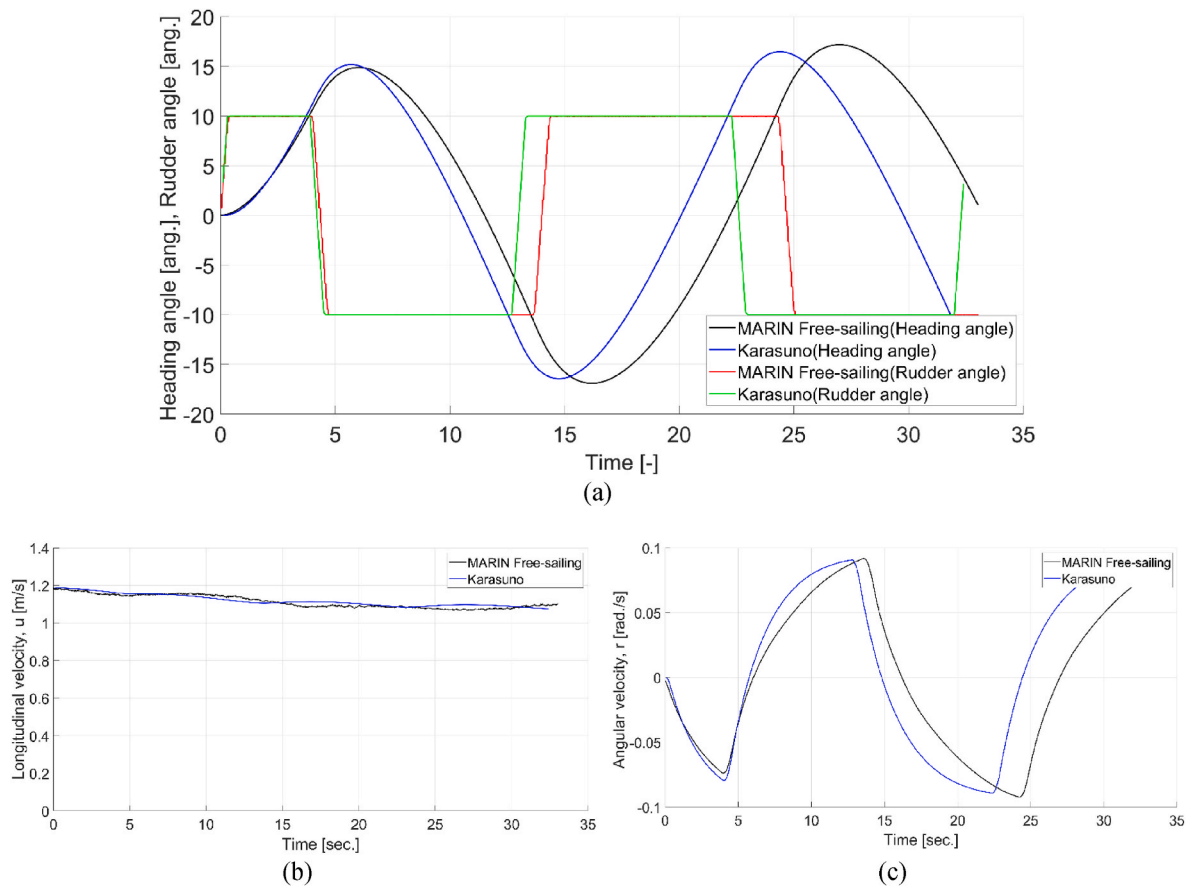


Fig. 23. Comparison results with free-sailing model test for horizontal 10/10 zig-zag manoeuvring: (a) Time histories for both heading angle and rudder angle, (b) Time histories for longitudinal velocity(u), (c) Time histories for yaw velocity(r).

Table 14

Comparison results for horizontal zig-zag manoeuvring motion.

	Overpelt et al. (2015)	Present	Difference $\left(\frac{\text{Overpelt} - \text{Present}}{\text{Overpelt}} \times 100\right)$ [%]
1st Overshoot angle	4.88	5.18	-6.15
2nd Overshoot angle	6.89	6.42	6.82

ship speed and yaw angular velocity during zig-zag maneuvering. As evident from Fig. 23, good agreement between the numerical simulation and model test is observed, which is also confirmed by the time series of speed drop and yaw angular velocity. For the horizontal 10/10 zig-zag maneuvering motion, the variation in heading angle is relatively small compared to turning maneuvering motion. This is attributed to the relatively minor differences in longitudinal force between the CFD calculations and model tests within the small angle range. Moreover, the comparison results for the 1st overshoot angle and 2nd overshoot angle presented in Table 14 agree with the model test within approximately $\pm 7.0\%$. Quantitatively, with a difference of approximately $\pm 0.4^\circ$, the agreement is considered satisfactory. However, there is a phase difference between the numerical simulation and model test, likely attributable to the accumulation of subtle differences in estimated forces and moments over time in the numerical simulation.

5.3. Vertical zig-zag manoeuvring simulation

The results of the vertical 10/10 zig-zag simulation were compared with those of the model test under identical conditions to the horizontal

zig-zag maneuvering motion and are presented in Fig. 24 and Table 15. Fig. 24 (a) compares the time series of the submarine's pitch angle and effective vertical rudder angle, while Fig. 24 (b) and (c) show the time series of changes in ship speed and vertical angular velocity during vertical zig-zag maneuvering. As observed from Fig. 24, there is relatively good agreement between the numerical simulation and model test, which is also evident from the speed drop time series. However, there is some discrepancy in the pitch angular velocity between the two, likely stemming from differences in the longitudinal force acting on the hull due to the rudder, as mentioned earlier. Similar to the horizontal 10/10 zig-zag maneuvering motion, the vertical motion also shows relatively good agreement between the CFD and model test for the same reasons. Moreover, the comparison results for the 1st overshoot angle and 2nd overshoot angle presented in Table 15 show differences of approximately -84.2% and 24.5% , respectively. Quantitatively, with differences of approximately -1.4° and -0.1° , respectively, the agreement is considered satisfactory. However, the phase difference between the numerical simulation and model test is presumed to be consistent with the explanation provided earlier.

6. Conclusions

As mentioned in the introduction, a new type of mathematical model that can cover a wider range of maneuvering motion is necessary at the initial design stage of the submarine. Therefore, this study developed a new type of physical based mathematical model that can be applied to particular maneuvering motion with a large drift angle and/or angle of attack in both the horizontal and vertical planes. In addition, this study focused on the validity of the proposed mathematical model and its applicability. To obtain the hydrodynamic forces and moments acting on

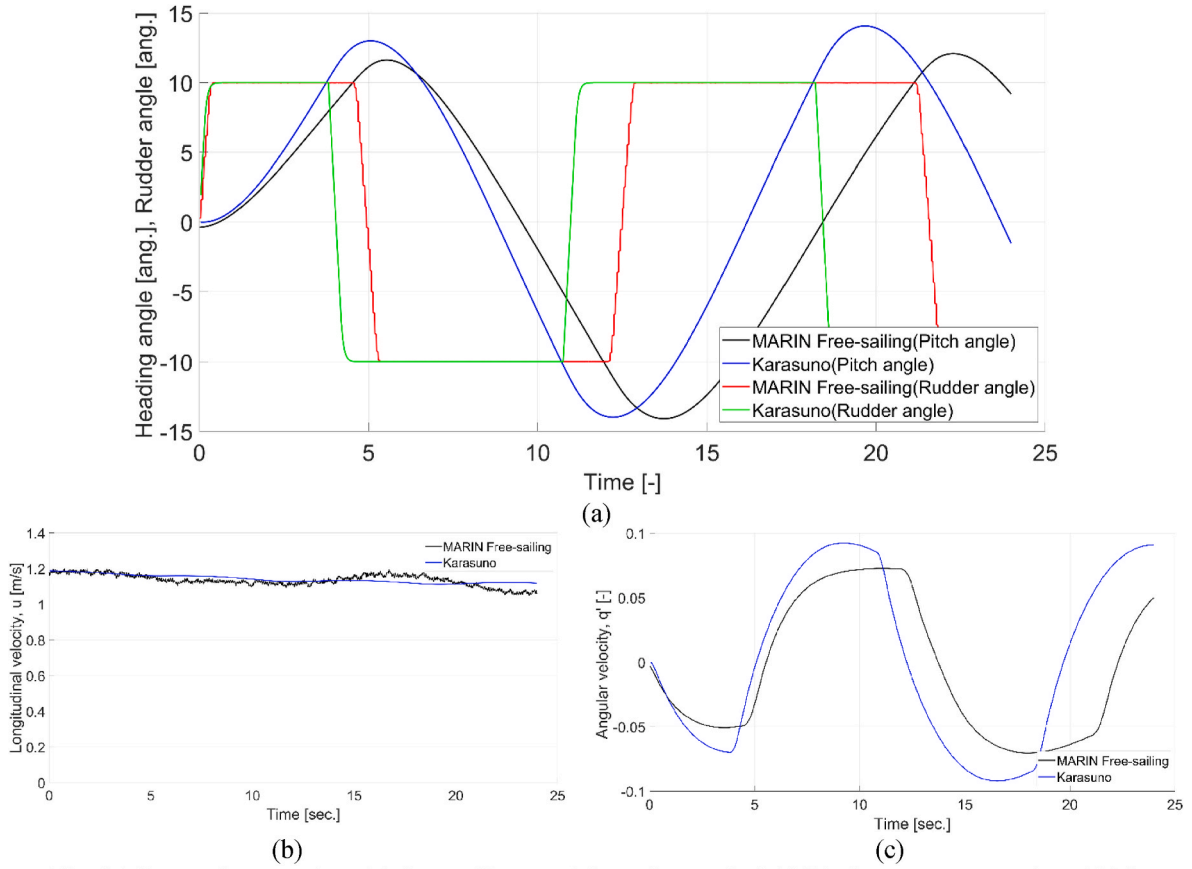


Fig. 24. Comparison results with free-sailing model test for vertical 10/10 zig-zag manoeuvring: (a) Time histories for both heading angle and rudder angle, (b) Time histories for longitudinal velocity(u), (c) Time histories for yaw velocity(r).

Table 15
Comparison results for horizontal zig-zag manoeuvring motion.

	Overpelt et al. (2015)	Present	Difference $\left(\frac{\text{Overpelt} - \text{Present}}{\text{Overpelt}} \times 100\right)$ [%]
1st Overshoot angle	1.62	3.0	-85.2%
2nd Overshoot angle	-4.08	-3.98	24.5%

the submarine in various manoeuvring motions such as horizontal static drift motion, vertical static angle of attack motion, and combined rotational motion with the drift angle, numerical simulations were performed instead of the model experiment because of the limitation of the experimental facilities. In addition, a series of validation studies for DARPA Suboff and MARIN BB2 were performed to guarantee the accuracy of the CFD results and compared with the published experimental results and other CFD results (Roddy, 1990; Etebari et al., 2008; Kwon et al., 2023).

Meanwhile, in order to obtain the hydrodynamic coefficients acting on a submarine in manoeuvring motions, a new method called Karasuno's physical based model, which has never been utilized with a submarine, was applied and compared with the traditional Taylor expansion mathematical model. In particular, the original Karasuno model was previously applied only to fishing vessels in 3DOF maneuvering motion on a horizontal plane. However, in this study, a method of combining each Karasuno submodel for horizontal motion and vertical motion was proposed for submarines, and a new Karasuno method for 6DOF motion was proposed as a result. In addition, in order to confirm the advantages of Karasuno model, CFD calculations for large drift and/

or angle of attack motion were performed.

From the results, it was confirmed that Karasuno's model can give reasonable results in a small range of drift angle and angle of attack (approximately $0^\circ \sim 50^\circ$) and it is found that Karasuno's model can give better results than the Taylor expansion mathematical model in large drift angle and angle of attack (approximately above 50°). Of course, the Taylor expansion mathematical model with higher order also can give better results but it is more complicated in terms of calculations.

After confirming the validity of Karasuno's physical based mathematical model at a force level, which is included in the equations of manoeuvring motion, a numerical simulation for some specific manoeuvring motions of a submarine were performed and compared with the experimental results by Overpelt et al. (2015). In the case of horizontal turning motion, the difference for advance of the turning motion is under a maximum of $\pm 0.5\%$, for transfer it is under a maximum of $\pm 11.0\%$, and for tactical diameter it is under a maximum of 13% . Also, in the case of horizontal 10/10 zig-zag motion, the difference for the 1st overshoot angle and for the 2nd overshoot angle are both under maximum of $\pm 7.0\%$. Quantitatively, with a difference of approximately $\pm 0.4^\circ$, the agreement is considered satisfactory. Finally, in the case of vertical 10/10 zig-zag motion, the difference for the 1st and 2nd overshoot angle is under maximum of 85% and 25% , respectively. It appears that the difference ratio is shown excessive but, in terms of the quantitative value (the difference of overshoot angle is small), it can be said that the accuracy is quite good.

Meanwhile, in order to confirm the applicability of the proposed manoeuvring mathematical model for submarines, further research should be conducted. In particular, the proposed physical based mathematical model has a limitation for determining parameters (c'_{vo} , x'_t and x'_l) that are related to the locations of the forces acting on the sub-

marine's hull during horizontal drift motion and vertical angle of attack motion. As mentioned in section 4.4.3, these parameters can be determined from the drift motion according to Karasuno et al. (1991, 1992). Also, this aspect is one of the advantages of Karasuno's model because it can help to reduce the experimental matrices. In contrast to the fishing vessel studied by Karasuno et al. (1991, 1992), in the case of submarines, the parameters for each motion (horizontal drift motion and vertical angle of attack motion) have different values and should be determined by using a trial and error method. To overcome this limitation, further study should be performed with various hull forms.

CRedit authorship contribution statement

Sungwook Lee: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jin-Hyeong Ahn:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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